

PICASSA: Platform for Interactive, Comprehensive and Accessible Sc-RNA Seq Analysis – Detailed manual



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Are you a **wet-lab biologist** struggling with single-cell RNA-sequencing (**scRNA-seq**) data, and feeling overwhelmed by coding and endless back-and-forth emails with bioinformatics cores?

PICASSA is designed for you!

It empowers you to take **full control of your data analysis**, no programming required. From raw data to publication-ready results, PICASSA streamlines the process, making single-cell analysis interactive, comprehensive, and accessible.

Introduction

This manual provides a step-by-step guide for conducting scRNA-seq analysis using PICASSA. Users should follow these guidelines to ensure accurate and efficient analysis. This **manual** offers a **detailed explanation** of how to perform each analysis step, navigate the application interface, and interpret outputs through practical explanations and examples. It is organized into two main chapters, Phase 1 and Phase 2, each divided into subchapters that follow the structure of the application interface to facilitate workflow navigation. The application interface also provides concise guidance via the “**Instructions**” dropdown menu at each stage of the workflow. These in-app instructions provide a **quick reference** for or users requiring brief operational guidance during analysis. Additional support is provided through a “**Quick Manual**”, available via the PICASSA [GitHub](#) repository and main webpage, which describes the **purpose** of each analysis step, the required inputs, and the corresponding outputs.

The PICASSA [GitHub](#) repository and main webpage also provide access to the application and to the **supplementary modules** “**Gene Conversion**” and “**Pathway Plotter**”. These modules **extend PICASSA functionality** for downstream analyses and are described in more detail in later sections of this manual.

PICASSA supports the analysis of either a **single sample** or **multiple samples**, including **replicates**, from both **human** and **mouse** data.

IMPORTANT NOTE ON NOMENCLATURE: Throughout the app, **experimental conditions** are referred to as **groups** (for example, control and treatment). When multiple **replicates** are present within the same experimental condition, these are referred to as **samples** (for example, the replicates control1 and control2 are considered samples within the control group).

See the table below for examples:

Experimental setup	Group(s) shown in PICASSA	Samples shown in PICASSA
--------------------	---------------------------	--------------------------

Only one sample (e.g., one control)	Control	-
One control sample + One treatment sample	Control Treatment	-
Two control replicates + Two treatment replicates	Control Treatment	Control1 Control2 Treatment1 Treatment2

The application provides various **customizable options**, allowing users to adjust parameters such as **colors**, graph **size**, font size, and label **orientation**. All generated **tables** can be **downloaded** in **Excel** format and **plots** exported as **PDF** files via their respective download buttons, while plots may also be saved as **PNG images** directly from the interface using the browser's image saving functionality (right-click and "**Save Image as**"). Tables and plots can be displayed and downloaded either per group or, when applicable, per sample via a dropdown menu. To ensure full traceability of the analysis workflow, users are strongly encouraged to **download the outputs as they are generated**, noting that some outputs may only become available for download at subsequent steps.

Please note that **inactive sessions are automatically terminated** after approximately **1 hour** to ensure efficient resource usage and a responsive experience for all users.

PHASE 1

Phase 1 includes the [standard preprocessing](#), dimensionality reduction, and clustering workflow to generate the initial Uniform Manifold Approximation and Projection (UMAP) plot and clustering results. This chapter of the manual provides a step-by-step guide to help users complete the procedure.

Note: Any executed **step** can be **revisited** and modified at any time. To do so, when in the desired step, adjust the parameters if needed, and click "**Run**" again. Clicking "**Run**" restarts the analysis from that step—even if the parameters remain unchanged—and all downstream steps are automatically refreshed to incorporate any changes introduced at that step. This ensures consistency throughout the workflow and is equivalent to continuing the analysis from that step onward.

To begin, click on "**Open Phase 1**" on the main interface.

Important notes before starting

If you have **not yet performed the analysis or wish to modify the filtering parameters** used in a previous analysis, the analysis should be run from the **beginning**. If the analysis has already been conducted up to the checkpoint at step 5, and no modification of the parameters in the previous steps is intended, the corresponding **seurat object** (.rds file) generated at that step must be loaded to **continue the analysis** using the refined dataset and maintain workflow consistency. Only .rds files generated by PICASSA are supported, as objects created from external sources may not be compatible with the pipeline.

Note: Your **analysis progress can only be saved once the workflow reaches step 5** (checkpoint), where it is stored as a **seurat** object. If you stop before this point, any completed analysis steps will not be saved and the analysis will need to be restarted.

Get started

Depending on the current stage of your analysis, you can either run the full workflow or begin from step 5 (checkpoint) by selecting the appropriate option presented below.

Under **“A) Start with your data”**, choose how you would like to proceed:

- **“Start a new analysis from scratch”**, if you are **beginning a new analysis or want to modify filtering parameters** of a previous analysis.
- **“Resume my previous analysis by loading my checkpoint.rds file (continue from step 5)”**, allows you to **load a previously downloaded seurat object** checkpoint file generated with PICASSA once the analysis reached step 5. After browsing and uploading the .rds file and selecting the organism used in the experiment, the workflow will resume from step 5, with the filtered and normalized data restored from the previous analysis stored in the checkpoint file. After clicking the initial **“Run”** button, the checkpoint data are read correctly. Clicking the subsequent **“Run”** button reproduces the outputs generated at the checkpoint step in the previous analysis before saving the .rds file, so that you can visualize them again and use them to support your decisions in the following steps. You can then **continue through the rest of the workflow**.

Note: An **example checkpoint file** is available for download through **“Download Example Data (PICASSA generated)”** allowing users to **test** file selection, upload, and workflow restoration from step 5.

Alternatively, under **“B) Start with a demo dataset”**, you can **explore Phase 1 without using your own data**. This section provides **example datasets** and includes **options analogous to those in section A:**

- **“Load example from start”**, where **experiment parameters** are **automatically defined** and the corresponding **input files** are **automatically loaded** in the **“Input Instructions”** tab; click **“Next”** to proceed through the workflow step by step.
- **“Load example from checkpoint (step 5)”**, where **experiment parameters** are **automatically defined** and the corresponding **input files** are **automatically loaded** in the **“Load checkpoint object”** tab, with cell and gene **filtering already completed**; click **“Run”** to perform **data normalization** and access the resulting outputs useful to support decisions in the following steps, as explained below in the subchapter **“5.Normalize - dimensionality reduction - compute CCS (checkpoint)”**. You can then continue through the rest of the workflow.

Start a new analysis from scratch

Input

1. Specify whether the experiment includes **replicates or not**.
2. Select the **organism** used in the experiment.
3. Click **“Let’s Start!”**.
4. Choose one of the **data types** below for the **input** specimen files:

- **10x – files** (barcodes.tsv.gz, features.tsv.gz and matrix.mtx.gz commonly found in the filtered_feature_bc_matrix folder) or **.H5 – file** (filtered_feature_bc_matrix.h5).

Created by cellranger (most common).

PICASSA preserves and integrates both Ensembl IDs and gene symbols, providing flexible gene representation and ensuring more comprehensive downstream analyses.

Note: When selecting one of these two data formats (if “no” was previously chosen for the presence of replicates), a corresponding example dataset is available for download based on the selected organism (“**Download example data (ZIP) – mouse**” or “**Download example data (ZIP) – human**”), allowing users to test file selection, uploads and sample/group naming, as described below, and optionally continue through the complete analysis workflow.

- **Gene-count matrix – file** (countMatrix.txt or countMatrix.tsv)

Created by other alignment/counting tools.

PICASSA attempts to detect and merge Ensembl ID and gene symbol columns automatically; if only one format is present, it will use that.

Note: All specimen files must be provided in the same format.

5. **Enter** the **specimen name** for which you intend to load the input files (i.e., Tumor).
6. “**Browse**” for the corresponding input file(s) and then “**Load**” it once the upload is completed.

Note: The **loading** can take **some time** so be patient and wait until the “**Load**” button becomes clickable and “**Upload complete**” appears.

7. Repeat steps 4-6 for each specimen..
8. After loading the files for all specimens, click “**Done! Next**”.

IMPORTANT: The **input files are just temporarily uploaded in PICASSA** server, once you exit the analysis they will be deleted.

9. Enter the group names in the desired order from the dropdown menu in the “**Enter group names**” box (or manually if replicates are present); this order will be reflected in the plots throughout the analysis.

Note: If only a single sample is loaded, assigning group names will not be required.

Note: If you are uploading replicates, each **sample name** must begin with its corresponding **group name**. Group names must be unique prefixes and should not be substrings of one another. Group names are case-insensitive, meaning that uppercase or lowercase letters do not matter.

See the **table** below for **examples**:

Samples	Groups	Reason
Correct examples:		
Control1, Control2 Disease1, Disease2	Control Disease	Group names are unique and appear clearly at the start of each sample
Control1, Control2 Disease1, Disease2 Treatment1, Treatment2	Control Disease Treatment	
Incorrect examples:		

1control, 2control 1disease, 2disease	Control Disease	Group name does not appear at the start of the sample name
Control1, Control2, Disease1, Disease2 Disease_Treatment1, Disease_Treatment2	Control Disease Disease_Treatment	The group names are ambiguous because one is the prefix of the other ("Disease" part)

10. Click **"Next"** to access the following step of the analysis.

1. Read data + cell filtering

- Under the **"Read data"** section, click **"Run"** to process the input data and access the following **outputs** useful for **identifying low-quality and/or potentially problematic cells and defining thresholds to filter them out**, all available in the **"QC before cell filtering"** panel:
 - Two **summary tables**:
 - "Summary of cells"**: Showing the number and percentage of cells before and after cell filtering.
 - "Details of cells"**: Comparing cell counts before and after applying thresholds for one or more quality control (QC) metrics (e.g. UMIs, genes, mitochondrial and ribosomal percent), along with the number of cells removed for each parameter.
Note: At this step these tables are not downloadable yet but they will be as soon as you will access the next panel.
 - Customizable **thresholds**, available in the **"Define Parameters"** sidebar, for the following **QC metrics** for cell filtering:
 - UMIs**: Number of Unique Molecular Identifiers (UMIs), which are equivalent to transcripts, detected in each cell.
Note: In addition, dataset-specific suggested limits are computed using the median to determine the **automatically prefilled** minimum and maximum **values** ($\text{median} \pm 5 \times \text{SD}$), providing a data-driven reference for setting the UMI cutoffs according to the dataset's variability. If the calculated minimum or maximum value falls outside the observed data range, it is adjusted to the nearest observed lowest or highest value, respectively. Visit this [page](#) for more details.
 - Genes**: Number of unique genes detected in each cell.
Note: Similar to the observation above, dataset-specific suggested limits are computed using the median to determine the **automatically prefilled** minimum and maximum **values** ($\text{median} \pm 5 \times \text{SD}$), providing guidance for selecting appropriate thresholds. If the calculated limits exceed the observed values, they are adjusted to the corresponding lowest or highest observed value. See [here](#) for more details.
 - Mitochondrial Percent**: Percentage of UMIs that map to the mitochondrial genome per cell.
Note: The maximum value is set by default to 5 for mouse data and 10 for human data as suggested by this [paper](#).

- **Ribosomal percent:** Percentage of UMIs that map to the ribosomal genome per cell.

Note: The **set values** for these thresholds determine the values in the tables described above and appear as **red lines** on the plots described in the next point.

IMPORTANT: The **prefilled values** are provided as **initial guidance** and should be **critically evaluated** using the diagnostic plots described below and **adjusted if necessary**, as optimal thresholds depend on the specific experimental dataset.

- Different **QC plots** are available in their corresponding subpanels to help you **examine** cell populations and **decide on thresholds** for filtering cells appropriate to the specific experiment:
 - **“Bar plots”:** It displays the number of cells by group.
 - **“Scatter plots”:** It shows the **number of genes** per cell (y-axis) versus the **number of UMIs** per cell (x-axis), colored by **mitochondrial percentage**, with each dot representing a cell. This plot helps assess the relationship between these parameters, as well as any dispersion of data points. Cells appearing in the bottom right quadrant have a high number of UMIs but only a few numbers of genes, which may indicate dying cells or a population of low-complexity cell types (e.g., erythroid cells). Click [here](#) for more details. Importantly, jointly visualizing these metrics may provide more reliable guidance for threshold selection than considering each QC metric in isolation, as is done in the distribution plots described below, thereby reducing the risk of misinterpretation of cellular signals (outlined [here](#)).
 - **“Density plots”:**
 - i. **UMI density plot:** It shows the distribution of the number of UMIs detected in each cell. Markedly higher UMI counts relative to the main distribution may indicate multiplets (i.e., a droplet in which multiple cells were captured during the microfluidic encapsulation). Applying a maximum UMI threshold may help remove cells with exceptionally high UMI counts that deviate substantially from the main distribution and are therefore more likely to represent multiplets. However, this approach does not replace dedicated doublet detection methods (available in the next step), as many doublets may still fall within the expected UMI range. Conversely, markedly lower UMI counts might correspond to droplets containing low-quality (dying, degraded or damaged) cells or ambient RNAs. It is important to keep in mind that different cell types and cells isolated from various species may differ in their total RNA content. For example, tumor samples often contain large tumor cells, which typically have higher RNA content, mixed with smaller tumor-infiltrating lymphocytes, which generally have lower RNA content.
 - ii. **Gene density plot:** It shows the distribution of the number of unique genes expressed per cell. Similar to the observation

with UMI counts, barcodes associated with unusually high numbers of genes might be multiplets, whereas barcodes with low number of genes might be droplets containing low-quality cells or ambient RNAs. As mentioned above for UMI counts, applying a maximum gene threshold may help exclude potential multiplets, but they should not be considered a substitute for dedicated doublet detection approaches (available in the next step). However, it should be noted that biological differences between cell types and species can affect this metric. For example, stem cells may express a higher number of unique genes.

- iii. **Mitochondrial percentage density plot:** It shows the distribution of the percentage of UMIs mapping to the mitochondrial genome per cell. Stressed and damaged cells often exhibit high mitochondrial content, possibly because of increased apoptosis (during which mitochondrial membranes lose integrity and mitochondrial RNA is released into the cytosol) and/or loss and degradation of cytoplasmic RNA from lysed cells and concomitant apparent enrichment of mitochondrial RNA that is protected by double membranes and is therefore among the last to degrade. This [paper](#) suggests that a threshold of 5% is optimal for mouse data and 10% for human data. However, it is important to take into consideration that mitochondrial transcript abundance varies with cell type and species. For example, metabolically active cells may express a higher number of mitochondrial genes, and filtering based on mitochondrial reads could introduce bias in the analysis.
 - iv. **Ribosomal percentage density plot:** It shows the distribution of the percentage of UMIs mapping to the ribosomal genome per cell. High ribosomal content is common in specific cell types and does not necessarily indicate poor cell quality. More info can be found [here](#).
 - **“Violin plots”:** Alternative visualization of **QC metric** distributions described in the density plots above.
2. In the **“Define Parameters”** sidebar, determine **thresholds** for **cell filtering** to **refine the dataset** by removing low-quality cells and/or potentially problematic cells while retaining high-quality cells for downstream analysis:
- Set **desired threshold** values for the relevant QC metric(s).

Note: If no value is provided for a threshold (left blank), the system will automatically assign the respective minimum or maximum value for that parameter in the dataset, resulting in no filtering being applied.

Note: When **threshold values are changed**, the **red lines** on the QC plots, which indicate the set threshold cutoffs, as well as the values of the **“Summary of cells”** and the **“Details of cells”** tables, **update automatically** to show how the adjusted thresholds affect the plots and cell statistics.

- Click “**Run**” in the sidebar to **apply the selected thresholds** for the different cell filtering parameters to the analysis. You will be automatically redirected to the “**QC after cell filtering**” panel where you will find the filtered QC plots in their corresponding subpanels and the relative tables now available for download.

Note: If, after running the cell filtering, a **change in thresholds** for one or more parameters is desired, return to the “**QC before cell filtering**” panel, set new values, and review how they affect the tables and plots before clicking the “**Run**” button. The tables and plots in the “**QC after cell filtering**” panel will then be **updated** accordingly.

- Click “**Next step**” to access the subsequent step of the analysis.

2.Doublet detection - DoubletFinder

Doublets occur when **two cells are mistakenly encapsulated in one droplet**, resulting in mixed gene expression profiles that can **distort downstream analyses**, including incorrect clustering, misclassification of cell types, and the identification of artificial cell populations. Therefore, their identification and potential removal requires careful consideration, as further elaborated in the following paragraphs.

This [paper](#) compared different computational doublet detection methods for scRNA data, and **DoubletFinder** was identified as the method with the **best detection accuracy** among the evaluated methods. When applied, DoubletFinder classifies predicted doublets into high-confidence and low-confidence groups based on the strength of the doublet prediction. **High confidence** doublets typically represent **heterotypic** doublets, where two different cell types are captured together. **Low confidence** doublets may represent **homotypic** doublets, which occur when cells of the same type are captured together and are more difficult to distinguish from singlets. The DoubletFinder [github page](#) and [paper](#) explain this in more depth.

Performing this step is **optional** and can be skipped via the “**Skip**” button. This option is provided for users who wish to quickly explore the dataset, as DoubletFinder can be **computationally intensive** and may require additional processing **time**.

Note: If this step is skipped, clicking the “**Skip**” button will directly display the “**Metrics data summary**” panel (described below) as the next output.

If you decide to run DoubletFinder, then:

- Select the **Chromium chemistry** used for library preparation from the corresponding dropdown menu. This information can be found in the Cell Ranger web summary under the “**Run Summary**” section, in the “**Chemistry**” row, and is required because each Chromium chemistry is associated with a different **expected multiplet rate** depending on the initial number of cells loaded for sequencing. DoubletFinder uses this parameter to estimate the expected number of doublets in the dataset and optimize doublet prediction accordingly. **Reference tables** containing the expected multiplet rates for each Chromium chemistry are available in the corresponding documentation pages below:

3' Chemistry

Single Cell 3' v2
Single Cell 3' v3
Single Cell 3' v3.1
Single Cell 3' v4
5' Chemistry
Single Cell 5' v1.1
Single Cell 5' v2
Single Cell 5' v3

- Click **“Run”** to generate diagnostic plots within the **“Plots”** panel, organized into the following subpanels, to help you **identify potential doublets** and assess **whether they should be removed**:

- **“UMAP overview”**: UMAP plot in which each dot represents a cell, displaying predicted single cells (singlet) and doublets, including both high-confidence (High, typically heterotypic) and low-confidence doublets (Low, potentially homotypic).
- **“UMAP group-level”**: The same UMAP plot as above split by group.
- **“Violin plots by group”**: Distribution of **QC metrics** per group. Each violin is splitted according to the DoubletFinder classification to allow QC metrics comparison between singlets and predicted doublets.

- Click **“1.Check cells summary”** to review the **detailed counts and percentages of singlets and predicted doublets** displayed in the **“Cells summary”** table within the **“Cells Summary”** panel, allowing visualization of how many cells are predicted as doublets.

Note: In the subchapter **“4.Cells Summary”** you can find a more detailed description of this table.

Note: At this step this table is not downloadable yet but it will be in step **“4.Cells summary”**.

- Select **“Only High Confidence”** or **“All (High + Low)”**, under the **“Remove Doublets or Not”** option, to visualize how each doublet removal strategy affects the UMAP plots and to determine which doublets should be excluded. If you do not wish to remove any doublets, select **“Do not remove”**.

Note: If **doublets** create small **clusters**, it is especially recommended to **remove** them.

Note: If doublets do not form clearly distinct clusters and/or if you are **uncertain** about removing them, it is recommended to proceed **without filtering** at this stage. You can **complete the analysis** and **evaluate** whether any unexpected patterns arise, such as biologically hybrid clusters or anomalies in cluster QC plots. If needed, you can then revisit this step, enable doublet removal, and **rerun** the analysis with a more informed decision.

- Click **“2.Run”** to **apply** the selected option to the analysis, **update** the violin plots accordingly and enable the **“3.Next step”** button.

Note: Running the doublet detection **multiple times** with the **same parameters** will produce the **same UMAP plot**, as PICASSA fixes the random seed internally for reproducibility. However, re-running the analysis after **changing upstream parameters** (e.g., cell filtering thresholds) may yield a **different UMAP plot** layout. This occurs because UMAP is a nonlinear dimensionality reduction algorithm whose structure depends on the specific set of input cells and features; modifying these

inputs alters the underlying neighborhood graph, leading to slight shifts or rearrangements in the resulting UMAP plot.

6. Click again **“1.Check cells summary”** to review the summary statistics after doublet removal displayed in the **“Final cells summary”** table within the **“Cells Summary”** panel. This helps you to assess the **effect** of the selected strategy and **revise** your choice if needed.

Note: In the subchapter **“4.Cells Summary”** you can find a more detailed description of this table.

Note: At this step this table is not downloadable yet but it will be in step **“4.Cells summary”**.

Note: If, after running DoubletFinder with a specific option selected, you wish to **change** the option, select a different option under **“Remove Doublets or Not”** to **update** the UMAP plots and **visualize** how the new selection affects them, then click **“2.Run”** to **apply** the selected option to the analysis, update the violin plots accordingly and enable the **“3.Next step”** button.

7. Continue with the analysis via the **“3.Next step”** button.

As described above, the **“Metrics data summary”** panel is displayed as the next output, independently of running doublet detection. If DoubletFinder is run, the panel will become available after clicking **“3.Next step”**. The table present in this panel provides **additional information** that can assist you in determining appropriate **thresholds** for gene filtering in the next step, **“3.Filter genes”**, where the use of these metrics for this purpose is described in more detail.

3.Filter genes

1. In the sidebar, **define** minimum **thresholds** for the following **gene filtering** parameters to **reduce technical noise** by filtering out genes with zero or very low counts or detected in only a small number of cells, which may skew downstream analyses, while **retaining biologically relevant genes** for downstream analysis:

- **“Include genes detected in at least this many cells”**, which specifies the minimum number of cells in which a gene must be expressed to be retained for downstream analysis. For example, if a threshold of 3 is set, only genes expressed in 3 or more cells will be retained for downstream analysis.

Note: A **prefilled value** of 3 is **automatically set** as a commonly used empirical default threshold and is provided as initial guidance.

IMPORTANT: This value should be **critically evaluated** and **adjusted** if necessary according to the specific characteristics of the dataset. The rows **“Percentage of genes expressed in N or fewer cells (including genes with zero counts in all cells)”** and **“Percentage of genes expressed in N or fewer cells (excluding genes with zero counts in all cells)”** in the **“Metrics data summary”** table of step **“2.Doublet detection -**

DoubletFinder” can be used to help determine an appropriate value for this parameter. The **“Search”** box can help you find these entries in the table.

- **“Get genes that express at least this number of counts across cells”**, which specifies the minimum total number of counts required for each gene across all cells to be retained for downstream analysis. For example, if a

threshold of 5 is set, only genes with at least 5 total counts across all cells will be retained for downstream analysis.

Note: A **prefilled value** of 5 is **automatically set** as a commonly used empirical default threshold and is provided as initial guidance.

IMPORTANT: This value should be **critically evaluated** and **adjusted** if necessary according to the specific characteristics of the dataset. The row **“Q0 of counts per gene”**, which represents the minimum number of counts detected for a gene, together with the rows **“Percentage of genes with total counts of N or fewer across all cells”** in the **“Metrics data summary”** table of step **“2. Doublet detection - DoubletFinder”** can be used to help determine an appropriate value for this parameter. The **“Search”** box can help you find these entries in the table.

Note: To avoid applying any filtering for a parameter, set its threshold value to 0, which disables the minimum requirement for that parameter and retains all genes with respect to that specific filtering criterion.

See the **example table** below, which shows gene counts across cells, for a practical illustration of how these two gene-filtering parameters work together when are set to their empirical default values described above:

	Cell1	Cell2	Cell3	Cell4	Cell5	Does the gene pass the 1st threshold?	Does the gene pass the 2nd threshold?	Is the gene retained for downstream analysis?
Gene1	1	2	2	0	0	Yes	Yes	Yes
Gene2	2	3	0	0	0	No	Yes	No
Gene3	1	1	1	1	1	Yes	Yes	Yes
Gene4	1	1	1	0	0	Yes	No	No
Gene5	1	1	0	0	0	No	No	No

Note: For more information on the importance of gene filtering and its impact on downstream analyses, see this [paper](#).

- Click **“Run”** to apply the selected thresholds for the gene filtering parameters to the analysis and display the following outputs in the **“QC after gene filtering”** panel:
 - “Gene filter summary”** table: Summarizing the number of genes before and after gene filtering.
 - “QC after gene filtering”: QC plots** analogous to those shown in the **“QC after cell filtering”** panel of step **“1.Read data + cell filtering”** from **“Start a new analysis from scratch”**, now **updated** to incorporate the changes introduced by gene filtering thresholds.
- Click **“Next step”** to access the **“Metrics data summary”** panel, where the corresponding table is displayed. This table is analogous to the one shown in the **“Metrics data summary”** panel of step **“2.Doublet detection - DoubletFinder”**, now **updated** to incorporate the changes introduced by gene filtering thresholds.
- Click **“Next step”** to access the next subsequent of the analysis.

4.Cells summary

If DoubletFinder was performed the same “**Cells summary**” and “**Final cells summary**” tables mentioned in step “**2.Doublet detection - DoubletFinder**” will be **displayed** and available for **download**. In both tables, rows correspond to groups, and columns provide information about cell counts and proportions. The first table contains the full set of metrics prior to doublet removal, while the second summarizes the data after applying the selected doublet filtering option. Percentages for “**Cells After Filtering**” are calculated relative to the total number of “**Cells Before Filtering**”. All remaining percentages are calculated relative to the total number of “**Cells After Filtering**”.

If DoubletFinder is not performed, only the “**Cells summary**” table is displayed without annotations for singlets and doublets.

1. Under the “**Cells summary**” panel, or under the “**Final cells summary**” panel if DoubletFinder was performed, click “**Download report**” to download a report **summarizing the executed steps** and corresponding **selected parameters** up to this step, to facilitate comparison of alternative runs, later review of the workflow, and accurate reporting of analysis settings in the Methods section of a publication, and to activate the “**Next step**” button allowing you to proceed with the analysis.
2. Continue with the analysis by clicking the “**Next step**” button.

5.Normalize - dimensionality reduction - compute CCS (checkpoint)

1. Click “**Run**” to perform **data normalization, dimensionality reduction** (using Principal Component Analysis, PCA, to compute Principal Components, PCs), and **cell cycle scoring** (CCS). These preprocessing steps **prepare the data for downstream analysis** and generate different **outputs** displayed in dedicated panels, as described below. Some of these outputs are useful for determining the number of PCs to use in downstream analyses (e.g., UMAP plot generation and clustering) and for evaluating sources of variation (e.g., cell cycle) to possibly regress out (i.e., reducing their influence on the data), guiding your decisions in the next steps of the workflow.
 - **Elbow plot**: It displays the **PCs** versus their respective **standard deviation**. This plot is useful for **evaluating the number of PCs to use for downstream analysis**. Because single-cell datasets contain expression measurements for thousands of genes per cell, **PCA** is used to **reduce** this high-dimensional data into a smaller number of variables, the **PCs**. PICASSA performs PCA using the default 3000 selected highly variable genes (**HVGs**), which capture the most informative sources of biological variation while reducing the influence of less informative or noisy genes. The PCs are computed as weighted combinations of expression values across these HVGs and summarize the main independent sources of variation across the cells in the dataset. PC1 explains the largest source of variation across cells, PC2 explains the next largest source of variation independent of PC1, and subsequent PCs capture progressively smaller patterns of variation. For example, in a heterogeneous sample, PC1 may separate immune from epithelial cells because this represents the strongest transcriptional difference

across cells. PC2 may then capture a different source of variation independent of cell type, such as cell cycle state or metabolic activity. Selecting an appropriate number of PCs for downstream analysis is essential:

- **If too few PCs** are chosen, critical **biological information** may be **lost**.
- **If too many PCs** are selected, **noise** may overshadow meaningful data.

Note: The **recommended number of PCs** to use for **downstream analyses** is **automatically computed** using data-driven criteria rather than user subjective visual inspection and appears as a **vertical line** on the elbow plot. If desired, users can **select a different** number of PCs for downstream analyses in step 7.

- **Variable genes plot:** It shows the **average expression** level of genes across cells (x-axis) versus their **variability** across cells (y-axis), with each dot representing a gene. **HVGs**, highlighted in red, exhibit greater cell-to-cell variation than expected based on their expression level and are therefore considered more informative for downstream analyses.
- **Gene loadings:** It displays the genes that **contribute** most strongly to the **first two PCs**. Genes with positive and negative loadings are associated with opposite expression patterns across cells (e.g., high expression in some cell types and low expression in others), and genes with larger absolute loading values have a stronger influence on the corresponding PC. These genes may help identify the biological signals driving variation in the dataset. For example, in a heterogeneous sample where PC1 separates immune from epithelial cells, immune cell marker genes may show strong positive loadings whereas epithelial cell marker genes may show strong negative loadings (or vice versa). This indicates that PC1 is primarily capturing transcriptional differences between these two cell populations.
- **Heatmaps PCs:** Heatmaps showing how the **expression of genes** associated with **each PC** varies across cells, for the automatically recommended number of PCs and ten additional subsequent PCs. Rows represent genes with the largest loading values for each PC, while columns represent cells, with colors indicating relative expression levels. Stronger and more structured color contrast, typically observed in earlier PCs, suggests biologically meaningful variation, whereas weaker or more diffuse contrast in later PCs may indicate increasing noise and diminishing biological signal. These plots help guide the selection of the number of PCs for downstream analyses.
- **Correlation plot:** It shows the **correlation** of each displayed **variable** (x-axis) to **different PCs**. Blue and red colors indicate positive and negative correlations, respectively. The larger the dot and the more intense its color, the higher its correlation. This plot helps identify whether biological or technical variables are associated with specific PCs and may therefore affect downstream analyses, indicating variables that may need to be **regressed out**. For example, strong correlations between early PCs and cell cycle scores ("**S.Score**", "**G2M.Score**" and "**CC.Difference**") may indicate that cell cycle effects could impact clustering and marker gene identification. In such cases, users may consider regressing out these variables in the

subsequent step to reduce the influence of cell cycle heterogeneity on PCs and downstream analyses. The displayed variables include “**S.Score**” and “**G2M.Score**”, which are described in more detail in the plot below, and “**CC.Difference**”, representing the difference between G2M and S phase scores and therefore capturing differences between proliferating cells in different cell cycle phases. Regressing out CC.Difference allows signals distinguishing non-cycling and cycling cells to be maintained while regressing out differences between cell cycle phases among proliferating cells, which are often not biologically relevant to the analysis.

- **PCA of cell cycle:** It displays **each cell** (dot) according to its position along the **first two PCs** (PC1 and PC2), colored by its **assigned cell cycle phase**. This plot is useful to assess the influence of cell cycle on PC1 and PC2. Cell cycle phase is determined by scoring each cell based on the expression of known S phase and G2/M phase marker genes. Cells expressing S or G2/M markers are assigned to the corresponding phase, while cells expressing neither strongly are labelled as G1. The G1 category includes both actively cycling G1 cells and non-cycling quiescent (G0) cells, since G1 and G0 cells lack clear and distinct transcriptional signatures. More information can be found [here](#) and [here](#). Clear separation or gradients between cell cycle phases across PC1 and PC2 indicate the presence of cell cycle effects, meaning that cell cycle contributes to these PCs. Since PC1 and PC2 capture major differences between cells and are used for downstream analyses, strong cell cycle effects in these principal components may influence clustering and marker gene identification and may therefore indicate the need for cell cycle regression.
- **PCA of cell cycle (split):** It is the same plot as the one above but split by cell-cycle group.

IMPORTANT: Depending on the experimental context, users may choose to **regress out** cell cycle effects. Regression is generally recommended when cell cycle acts as a confounding factor and interferes with downstream analyses such as clustering or marker gene identification. In such cases, regressing out cell cycle scores may help reduce cell cycle-driven clustering and improve the identification of biologically relevant cell populations. However, if proliferation itself is biologically relevant to the experiment (e.g., cancer, stem cells, or immune activation), preserving cell cycle effects may be preferable and regression should be avoided. Therefore, the decision to apply cell cycle regression should be guided by both the biological context and the observed clustering patterns. **If there is uncertainty** about the results, users can resume the analysis from the checkpoint, perform the analysis again with or without cell cycle regression, and compare the results to assess the effect of cell cycle on downstream analyses and better determine whether cell cycle regression should be applied.

2. Click “**Download seurat object**” to download an .rds file containing the **current analysis state** and to activate the “**Next step**” button to immediately move on with the analysis. This file acts as a **checkpoint** and can be **reloaded** at any time to continue the analysis from this step, if you are not continuing the analysis immediately, by selecting “**Resume my previous analysis by loading my checkpoint.rds file (continue from step 5)**” in the “**Get started**” section under

“Open Phase 1”, as explained in the above “Get started” subchapter of the manual.

6. Regression

1. Using the outputs generated in the previous step to guide your decision, under “Regress Out Variables?” in the sidebar, select:
 - **No**, if you don't want to regress out any variable. You will be automatically redirected to the next step of the analysis.
 - **Yes**, if you want to regress out one or more variables selected from the dropdown menu. After **selecting** one or more variables, click “Run” to regress them out. The **same plots** generated in the previous step will then be displayed again in the corresponding panels to **confirm the effectiveness of the regression** and to **reassess the recommended number of PCs** to be used in downstream analyses, which may change following regression. Continue the analysis via the “Next step” button.

7. Determine PCs

1. Click “Run” to access UMAP plots for each group and a combined UMAP colored by group, in the “UMAP of groups” panel, generated using the recommended number of PCs computed in step 5, automatically pre-filled in the “Minimum PCs” box under “How many PCs to include” in the sidebar.

These plots provide **guidance** for the decision in the next step, where the user has to decide between data merging and data integration to ensure optimal clustering and downstream analysis:

- **Merging**: This option combines the datasets **without applying any batch correction**. In other words, the original expression structure is preserved more directly. It is the preferred option when cells from different groups are **uniformly distributed** in the UMAP plot.
- **Integrating**: This option applies **batch correction** – an alignment procedure that modifies the data representation. It is recommended when cells from different groups are **clearly distributed into separate regions** of the UMAP plot, which is commonly indicative of batch effects. Integrative analysis aligns shared cell types and states across datasets to mitigate batch effects arising from factors such as different tissues, time points, sequencing runs performed at different times, or differences in sequencing technologies, thereby improving comparability and increasing statistical power. Additional information regarding Seurat integration can be found in the [Seurat integration tutorial](#), which includes an example of when integration may be appropriate.

Note: The **number of PCs** to use in downstream analysis can be **modified** by entering a different value in the “Minimum PCs” box.

2. Click “Next step” to access the following step of the analysis.

8. Multiple plots/CCS

1. Supported by the explanations provided in the previous step, choose whether to “Merge” or “Integrate” your data in the sidebar.

If **integration** is selected, a list of available integration methods will be displayed. This step can be performed **multiple times** using different methods, as there is no universally optimal approach; the most suitable method depends on the characteristics of the specific dataset. However, integration can be **computationally intensive** and should be approached with care. It is recommended to begin with the **default** method (**Harmony**) and only explore **alternative approaches**, such as CCA or RPCA, if the **overlap** between conditions is **not satisfactory**. Repeatedly running integration is discouraged, as it can consume substantial memory resources and may lead to session instability or crashes. Below is a summary of the available integration methods:

Integration Methods

Users can select different methods for batch correction:

- **Harmony**: Iterative correction in PCA space. A fast method that effectively corrects strong batch effects but may require parameter tuning to avoid overcorrection.
- **CCA (Canonical Correlation Analysis)**: Finds correlated features between datasets. Captures shared biological variation across datasets but can be computationally intensive for large datasets.
- **RPCA (Reciprocal PCA)**: PCA-based integration with MNN correction. Fast and memory-efficient for large datasets, though it may perform less well when batch effects are very strong.

Note: If integration does **not** produce a **meaningful improvement** in group or sample **distributions**, **merging** the datasets may be **more appropriate**. This is because **integration** should **not** be treated as a **mandatory** preprocessing step, as unnecessary or overly aggressive batch correction can overcorrect the data, reduce biological variability and mask true differences between groups by removing variation that is biological rather than technical. This concern has been highlighted in several benchmarking studies including [study1](#) and [study2](#), showing that integration methods differ in how they balance batch removal against biological conservation, and that some methods can introduce detectable artifacts when correction is applied unnecessarily. In this context, PICASSA uses **Harmony** as the **default** starting method because it has performed well in a benchmarking [study](#), and is generally suitable as a first-pass integration method.

2. In the sidebar, **define values** for the following clustering parameters, which **determine** how cells are divided into **different clusters**:
 - **K parameter**: It defines the **number of neighboring** cells considered for each cell during clustering. Values ranging from 5 to 100 can be entered in the box under “**Enter k values (comma separated and no spaces)**” as a comma-separated list without spaces (for example: 10,20,30).

Note: **Prefilled values** of 10 and 20 are **automatically provided** by default as empirical initial values based on commonly used settings, with 20 being a widely used default choice for clustering analyses. If none of the resulting clustering configurations are satisfactory, testing **additional or alternative** k values is recommended.

- **Resolution range**: It controls **cluster granularity**, with higher values producing more clusters. The accepted range in PICASSA is 0.1 to 2. The minimum value can be entered in the box under “**Start resolution value**”, and the maximum value can be entered in the box under “**End resolution value**”. Providing both a starting and

an ending value allows testing a range of resolutions in increments of 0.1 intervals within the selected range. For example, entering 0.1 as starting point and 1 as the ending value will test resolutions from 0.1 to 1 in increments of 0.1 (0.1, 0.2, 0.3, ..., 1.0).

Note: Prefilled starting and ending resolution **values** of 0.1 and 1, respectively, are **automatically provided** by default as initial guidance based on commonly used empirical values for clustering analyses. If none of the resulting clustering configurations are satisfactory, testing **additional or alternative** resolution values is recommended.

3. Click **“Run”** to generate and access different **outputs**, described below, displayed in dedicated subpanels, listed below, under the **“Multiple Plots”** panel, which are useful for guiding the **selection of final clustering** parameters:
 - **UMAP of groups:** It shows UMAP plots for each group, together with a second combined UMAP colored by group.
 - **Clustering trees:** Clustering trees are generated **for each K parameter** and resolution value inserted above and illustrate **how clusters evolve** across **different resolution** values. The graph can be interpreted as follows:
 - **Horizontal levels:** Each horizontal level corresponds to a different **clustering resolution**.
 - **Nodes (circles):** Each node represents a **cluster** at a specific resolution. Node size reflects the number of cells within that cluster.
 - **Vertical paths:** Each vertical tree represents a cluster identified at the lowest resolution (top horizontal level) and tracks how that cluster may split, merge, or remain stable across **progressively higher resolutions** (horizontal levels below).
 - **Edges (lines/arrows):** Connect **related clusters across consecutive resolutions**, showing how cells are reassigned as resolution increases. Branching arrows indicate that a cluster splits into multiple subclusters, while a single continuous arrow indicates that the cluster remains relatively stable across resolutions. Brighter and thicker edges indicate a larger number of shared cells between connected clusters.
 - **Multiple UMAPs:** Display the different UMAP plots with **cells colored** according to the **cluster assignments** generated using the selected combinations of K parameter and resolution values. These plots allow comparison of clustering results across parameter settings and help **identify the most appropriate clustering configuration** for downstream analysis.
4. Click **“Next step”** to access the **“Cell cycle effects”** panel, which provides different **outputs** useful for **cell cycle** evaluation, organized into the following subpanels:
 - **“UMAP phase overview”:** It displays the UMAP plot with cells colored according to their assigned cell cycle phase.
 - **“Group-level plots”:** It provides complementary visualizations for evaluating cell cycle phase distributions and their relationship with clustering, including:
 - UMAP plots displayed by group and colored by cell cycle phase.
 - Separate UMAP plots showing only cells assigned to the G1, G2M or S phases.
 - The percentage of cells assigned to each cell cycle phase (G1, G2M or S) within each group.

Note: This analysis provides an additional evaluation of the impact of cell cycle on clustering, alongside the analyses described in the subchapter “**5. Normalize - dimensionality reduction - compute CCS (checkpoint)**”. If cells belonging to the same cell cycle phase cluster together, the cell cycle may represent a major source of variation influencing the clustering results. In this case, **regression** of cell cycle effects may be considered depending on the experimental design and biological question. However, if cell cycle state is biologically relevant to the study, retaining this variation may be preferable, as already discussed in subchapter 5.

5. Access to the next step of the analysis by clicking the “**Next step**” button.

9. Final plots/marker genes

1. Based on the different UMAP plots generated in the previous step within the “**Multiple UMAPs**” subpanel, **select** in the sidebar under “**Final Plots**” the **k parameter** and **resolution** corresponding to the **preferred clustering solution**, among those previously tested, using the dedicated dropdown menus.
Note: It is generally recommended to select k and resolution values that produce the **highest number** and **clearly separated** clusters while maintaining a **clean UMAP** structure. Although higher resolution values may generate more clusters, excessively high resolutions can result in overlapping or poorly separated clusters, making the clustering difficult to interpret biologically. Therefore, preference should be given to clustering configurations where clusters remain distinct and well separated, as this facilitates downstream analyses such as marker gene identification and enables more reliable decisions in Phase 2 regarding cluster merging, removal, or further subclustering.
2. In the sidebar under “**Marker genes – Between clusters**”, specify the desired **thresholds** for the **parameters** used to detect **marker genes**. Marker genes are genes differentially expressed between clusters (each cluster versus all others) that help **define cluster-specific cell identities** and are therefore used for UMAP **clusters annotation**, i.e., the assignment of biological identities to transcriptionally distinct cell clusters. These parameters collectively control the **stringency** of marker gene identification, as a gene must satisfy all specified thresholds to be considered a marker gene, and include the following:
 - **Log2FC threshold:** It defines the minimum **difference in gene expression** between cells in a specific cluster and the remaining cells in all other clusters required for a gene to be considered a marker gene. A value of $\log_2FC = |1|$ means that the expression change is double or half (fold change $[FC] = 2$ or 0.5). For example, if a gene has an average expression of 8 in cluster 1 and 4 across all other clusters, the FC is 2 ($8/4$) and the corresponding $\log_2FC$ is 1. Conversely, if the average expression is 4 in one cluster and 8 across all other clusters, the FC is 0.5 ($4/8$) and the corresponding $\log_2FC$ is -1 . The FC can be calculated as 2 raised to the power of the $\log_2FC$ value. For example, if $\log_2FC = 1$, then $FC = 2^1 = 2$, whereas if $\log_2FC = -1$, then $FC = 2^{-1} = 0.5$. If the Log2FC threshold is set to 0.58, a gene with an average expression of 6 in cluster 1 and 5 across all other clusters would not pass the threshold since the FC is 1.2 ($6/5$) and the corresponding $\log_2FC$ is approximately 0.26.

- **Minimum pct threshold:** It defines the **minimum percentage (pct)** of cells that must **express a gene in at least one of the two cell populations** (cluster under consideration-pct.1 or all other clusters-pct.2) being compared. For example, if the threshold is set to 0.2 (20%), a gene expressed in only 19% of the cells in cluster 1 and 15% of the cells in all other clusters will not pass the threshold, because it does not reach the minimum required expression fraction in either cell population being compared. This threshold **prioritizes** genes that are **widely expressed** across cells of a cluster, consistent with the concept of **canonical marker genes**, which are expected to characterize most cells within a given cell type rather than only a small subpopulation. For example, although a gene expressed in only 10% of the cells of a cluster may appear significantly enriched compared with other populations, it would not represent an informative marker for defining the identity of the entire cluster because it characterizes only a minor subset of cells.

Note: Because this parameter is satisfied if a gene is highly detected in either the cluster under consideration or the remaining comparison cells, candidate markers should additionally be inspected to ensure that they are more broadly expressed in the cluster under consideration than in the remaining clusters. For example, with min.pct = 0.15, a gene such as *S100A8* expressed in only 12% of macrophages but in 78% of the comparison population (e.g., neutrophils) would still satisfy the threshold because the required expression fraction is reached in one of the two populations being compared. However, such a gene would not represent a biologically meaningful macrophage marker because it is preferentially associated with the comparison populations rather than with the cluster under consideration.

Note: Genes with **more restricted expression** patterns are not necessarily biologically irrelevant, as they may reflect **distinct cellular subpopulations** (i.e., an activation state of that population) rather than canonical population-wide markers. Such genes may therefore still be valuable for identifying biologically meaningful cell types. For instance, with a minimum percentage threshold of 0.2 (20%), a gene such as *IFNG* expressed in 18% of T cells and only 2% of the remaining cell populations would fail the threshold because it is expressed in too few cells within the cluster under consideration. Although such a gene would not qualify as a population-wide marker gene because it is not broadly expressed across the entire cluster, it may still identify a biologically meaningful activated T-cell subpopulation and could justify further subclustering analyses for more targeted downstream analysis (discussed further in Phase 2).

- **Minimum difference of pcts threshold:** It defines the **minimum difference in the percentage of cells expressing a gene between the two cell populations** ($\text{absolute_difference}(\text{pct.1} - \text{pct.2})$) being compared. For example, if the threshold is set to 0.15 (15%), a gene expressed in 90% of the cells in cluster 1 and 80% of the cells in all other clusters will not pass the threshold, because the difference in expression fraction between the two cell populations is only 10%, which is below the required threshold. This threshold **prioritizes** genes showing **substantial differences** in detection frequency between cell populations, consistent with the concept of marker genes, which

are expected to be broadly expressed across cells of a cluster while remaining relatively restricted in other cell populations, thereby uniquely defining a specific cell population.

Note: Because this parameter is satisfied whenever the absolute difference in the fraction of cells expressing a gene between the population under consideration and the remaining comparison cells exceeds the specified threshold, irrespective of whether expression is higher in the target population or in the comparison cells, additional inspection is necessary to ensure that candidate markers are preferentially expressed in the cluster of interest rather than broadly expressed across multiple cell populations. For example, with a minimum difference threshold of 0.15 (15%), a gene such as *S100A8* expressed in 20% of macrophages but in 70% of other myeloid cell populations would still satisfy the threshold because the difference in expression fraction between the two populations exceeds the required cutoff. However, such a gene would not represent a biologically meaningful macrophage marker because it is expressed in a larger fraction of other myeloid cell populations than macrophages.

Note: Broadly shared lineage markers are not necessarily biologically uninformative, despite potentially failing stringent specificity thresholds, as they may identify larger cellular lineages rather than uniquely distinguishing closely related cell populations. For instance, with minimum difference of pct threshold of 0.15 (15%), a gene such as *CD11b* expressed in 99% of macrophages and 98% of other myeloid cell populations would fail the threshold because the difference in detection frequency is only 1%, despite representing an important broad myeloid/immune lineage marker. Biologically, *CD11b* is highly informative for identifying myeloid cells but is not useful for distinguishing macrophages from other myeloid populations because it is broadly shared between both populations.

- **Adjusted p-value threshold:** It defines the **statistical significance** threshold used to identify marker genes. Only genes with an adjusted p-value below this threshold will be included in the marker gene list. Lower threshold values increase stringency and reduce the likelihood of false positives.

Note: Commonly used threshold **values** are **automatically prefilled** to provide initial guidance but can be **adjusted** depending on the desired level of stringency and the resulting marker gene output. For instance, if a well-known marker gene for the cell type represented by cluster 1 is not included in the marker gene list because its log2FC value is 0.5 while the selected threshold is 0.58, the user may consider lowering the log2FC threshold. However, it is important to keep in mind that **marker genes** are intended to **distinguish cells belonging to one cluster from cells in other clusters** (i.e., expected to show clear transcriptional differences and potentially represent different cell types or states). Consequently, lowering the threshold also reduces stringency and may result in genes being classified as marker genes despite showing only small expression differences between clusters.

Note: Setting thresholds for these parameters helps prioritize **robust population-defining marker genes** by favoring genes that are positively enriched in the cluster under consideration ($\text{avg_log2FC} > 0$), broadly expressed across cells within that cluster (high pct.1), substantially more frequently expressed in the target cluster than in the remaining cell populations (high pct.1 – pct.2) and statistically

significant. Collectively, these criteria identify markers that are enriched, broadly prevalent, cluster-specific, and statistically robust.

3. Click **“Run”** to generate and access the following final Phase 1 outputs, available in the **“Save some data”** panel:
 - A **seurat object**, downloadable via the **“Download seurat object”** button, which contains all processed data, metadata, and analysis results. Saving this object is required to activate the **“Next step”** button and continue in the workflow. The saved object can **subsequently be loaded in Phase 2 to continue the analysis** from this stage and ensure reproducibility of the performed analyses.
 - **Distribution tables**, downloadable via the **“Download distribution tables”** button, are organized into separate sheets containing the following information and corresponding sheet names:
 - Cell **counts** and relative **percentages by cluster and group**, available in the **“Cluster_Dist_Norm_byGroups”** sheet.
 - Percentage of **cells** from each group **within each cluster**, available in the **“Group_Cluster_Percentages”** sheet.
 - Summarizes the number and proportion of cells from each group within every final cluster. Fisher’s exact test is used to assess whether each **cluster is significantly enriched or depleted** in a given group relative to the overall distribution of cells across all groups, beyond what would be expected by chance, available in the **“Group_Cluster_Statistics”** sheet.
 - **Number** of downregulated and upregulated **marker genes** for each cluster, available in the **“MarkerGenes_Numbers”** sheet.
 - Cell **counts** and **percentages** for each **cell cycle phase within each group**, available in the **“CCE_Across_Groups”** sheet.
 - Cell **counts** and **percentages** for each **cell cycle phase within each cluster**, available in the **“CCE_Across_Clusters”** sheet.
 - **Marker gene tables**, downloadable via the **“Download unfiltered marker genes”** and **“Download filtered marker genes”** buttons. Both tables contain the same type of information; however, the **filtered** table includes only **genes** that **pass the selected marker gene thresholds**, whereas the **unfiltered** table contains all tested genes generated using Seurat’s **default parameters** except that **no minimum expression difference** cutoff is used. This allows **retention of genes with small or no expression differences** that are otherwise excluded by the default threshold, while the remaining default parameters are preserved to **help limit noise** in downstream analyses. The unfiltered table can be useful for investigating why specific genes did not qualify as marker genes and for evaluating whether different threshold settings may be more appropriate. The information in these tables is organized into separate sheets. The first sheet serves as a guide to the remaining sheets and indicates where the marker gene information for each cluster is located. The **“AllClusters”** sheet contains marker gene information for all clusters together, whereas the remaining sheets contain the same information split by individual clusters. Each sheet contains one row per gene and includes the following columns:
 - **gene**, which includes gene name - EnsemblID.

- **p_val**, which includes raw p-value.
- **avg_log2FC**, which includes the average log2FC in gene expression between the cluster under consideration and all other clusters combined. Positive values indicate higher expression in the cluster under consideration, whereas negative values indicate higher expression across the other clusters when considered together..
- **pct.1**, which includes the **percentage of cells expressing the gene** in the specific **cluster**.
- **pct.2**, which includes the **percentage of cells expressing the gene** in **all clusters** excluding the one under consideration.
- **p_val_adj**, which includes the adjusted p-value
- **cluster**, which identifies the cluster under consideration.
- **EnsemblIDs**, which includes the EnsemblID of the gene under consideration.
- **GeneNames**, which includes the gene name of the gene under consideration.
- **avg_exp1**, which includes the **average gene expression** in the **cluster under consideration**.
- **avg_exp2**, which includes the **average gene expression** in **all clusters** excluding the one under consideration.
- **N_cells_1**, which includes the **number of cells** in the **cluster under consideration**.
- **N_cells_2**, which includes the **number of cells** in **all clusters** excluding the one under consideration.
- **Expression**, which indicates the **direction of differential expression** for the gene relative to the cluster under consideration (“UP” or “DOWN”). In the unfiltered tables, genes that do not meet the selected marker gene thresholds are labeled as “NO”.

Note: **Average gene expression values** are **important** when interpreting marker genes, since log2FC reflects relative differences in expression between clusters. For example, consider two large macrophage clusters, one expressing CD11b at high levels and the other at lower levels, together with a small T-cell cluster not expressing CD11b. In this case, the macrophage cluster with lower CD11b expression may show a negative log2FC for CD11b when compared with all other clusters, simply because the second macrophage cluster expresses the gene even more strongly. However, the average expression values would still show that CD11b is highly expressed in both macrophage clusters compared with the T-cell cluster. Therefore, average expression information provides important biological context for gene expression evaluation and **marker gene interpretation**.

Note: If the analysis is rerun using different thresholds for marker gene identification, the unfiltered data table will still appear as already downloaded since it is unaffected by threshold changes.

- **Ranked gene tables for GSEA**, downloadable via the “**Download clusters prerank tables for GSEA app**” button. The information in these tables is organized into separate sheets, with two sheets generated for each cluster. Each sheet contains one row per gene and includes the following columns:
 - **gene**, which includes gene name - EnsemblID.

- **log2fc_jitter / RankMetric**, which includes the ranking value assigned to the gene depending on the selected ranking metric approach.
- **GeneNames**, which includes the gene name of the gene under consideration.
- **EnsemblIDs**, which includes the EnsemblID of the gene under consideration.

The two sheets generated for each cluster correspond to the following two different **gene-ranking strategies** used for Gene Set Enrichment Analysis (GSEA):

- **Avglog2fc_CI_X**: Genes **ranked** according to the **average log2FC**. **Note:** A small random noise value (jitter) is added to the computed log2FC to avoid tied ranking values and ensure stable gene ordering. This is why small differences may occasionally be observed between the log2FC values reported here and those reported in the unfiltered marker gene list.
- **RankMetric_CI_X**: Genes **ranked** according to the **statistical significance** of their differential expression. More statistically significant genes receive higher absolute ranking values, whereas positive and negative values indicate whether the gene is upregulated or downregulated in the cluster.

Note: Both ranking strategies are widely used in GSEA analyses and can be tested and compared to determine which approach provides the preferred enrichment results for the given dataset.

Note: The two ranking strategies ensure stable ordering of genes within preranked lists and compatibility with downstream GSEA tools. The exported files can be easily **converted into .rnk** format for use in **GSEA** workflows. More details on GSEA prerank using the app can be found [here](#) and on how to create the .rnk file [here](#). The user has the possibility to **run GSEA analysis also in Phase 2 of PICASSA**. However, running GSEA externally using the exported files allows greater flexibility in parameter selection and analysis settings, whereas in Phase 2 GSEA is performed through the integrated R function using the default settings recommended by the GSEA guidelines.

Note: If the analysis is rerun using different thresholds for marker gene identification, the “**Download clusters prerank tables for GSEA app**” will still appear as already downloaded since they are generated from the complete unfiltered marker gene table and are therefore unaffected by threshold changes.

Note: Unless otherwise specified, the term “**marker genes**” refers to the **filtered marker genes** from this point onward.

4. Click “**Next step**” to generate and display the “**Final plots**” panel containing different outputs organized into the following subpanels:
 - “**Main plots**”, which contains:
 - **Elbow plot** highlighting the **final chosen number of PCs** used for the analysis with a **vertical line**.
 - **Final UMAP plot** showing the **final clustering solution** generated using the selected k parameter and resolution values. This UMAP is identical to the one displayed in step 8 corresponding to the selected clustering solution tested by the application.

- **Stacked bar plot** showing the **distribution of marker genes** across clusters, separated into upregulated and downregulated genes according to the selected marker gene thresholds.
 - **UMAP plot** colored by cell cycle phase.
 - **Stacked bar plot** showing the relative percentage **composition** of **cell cycle phases** across **clusters**, allowing comparison of cell cycle phase contributions across clusters..
 - **“Group-level plots”**, which contains:
 - **UMAP plot split by group** and **colored by cell cycle** phase.
 - **UMAP plot split by cell cycle** phase, with cells **colored by group**.
 - **Stacked bar plot** showing the relative percentage **composition** of **each cell cycle phase** across **groups**, allowing comparison of cell cycle phase contributions across groups.
 - **UMAP plot split by group**, with cells **colored** according to their final **cluster** assignment.
 - **UMAP plot split by group** and **colored by** their **group** of origin.
 - **Stacked bar plot** showing the relative percentage **composition** of **each cluster by group**, allowing comparison of group contributions across clusters.
 - **Stacked bar plot** showing the relative percentage **composition** of **each group by cluster**, allowing comparison of cluster contributions across groups.
5. Click **“Next step”** to generate and display the **“Cluster QC”** panel, which contains multiple subpanels named after the displayed QC metric. Each subpanel includes a **feature plot** showing the distribution of the metric **across the UMAP** plots and a **violin plot** showing the **metric** distribution **by cluster**. These plots can help determine whether **more stringent** quality control filtering is needed.
 6. Click **“Next step”** to generate and access the **“Top 10 marker genes summary”** panel. A **table** displaying the top 10 marker genes for each cluster is provided, **ranked** primarily by decreasing average **log2FC** (highest FC first) and, in cases of equal FC values, **secondarily** by increasing **adjusted p-value** (smallest adjusted p-values first).
 7. Click **“Next step”** to generate and display the **“Top 5 marker genes feature plots”** panel, which displays **feature plots** showing the distribution of **gene expression across the UMAP** plot for the top five marker genes for **each cluster**, ranked as described above and displayed by cluster.

Note: Use the **“Select cluster to show”** dropdown menu to display the cluster of interest in this panel and in the plots generated in steps 8 and 9.

Note: These plots, together with those generated in Steps 8 and 9, support the visualization and interpretation of gene expression patterns.
 8. Click **“Next step”** to generate and display the **“Top 5 marker genes violin plots”** panel, which displays **violin plots** showing the **expression** of the top five marker genes for **each cluster**, ranked as described above and displayed by cluster.
 9. Click **“Next step”** to generate and display the **“Top 5 marker genes dot plots”** panel, which displays **dot plots** showing the **average expression** of the top five marker genes for **each cluster**, ranked as described above and displayed by cluster.
 10. Click **“Next step”** to access the following analysis step.

Note: At this stage, selecting k parameter and resolution values that produce the highest number of clearly separated clusters while maintaining a clean UMAP structure provides an initial guideline for choosing a clustering solution. However, this selection is still primarily based on the overall visual structure of the UMAP, and the optimal clustering solution is not always immediately obvious from the visualization alone. Therefore, the **gene expression profiles** and marker genes of each cluster should always be **critically evaluated** before deciding on the final clustering solution. Depending on these results, users may either proceed to **Phase 2** for **additional refinement** (for example, merging, removing, or subclustering specific populations) or **return to the checkpoint generated at step 5 and rerun steps 8-9 using different k parameter and resolution combinations** if a more suitable clustering configuration is desired.

10.Report

1. **Download** the report table via the “**Download report**” button to retain a comprehensive **record of all analysis** steps performed in Phase 1 and the corresponding **selected parameters**, summarized in the “**Parameter Name**” and “**Parameter Value**” columns, respectively. This facilitates reproducibility, supports iterative analysis with alternative parameters, and provides a detailed reference for accurately reporting the methods and analysis settings in the methods section of a publication.

Note: The **values** in the “**Parameter Value**” column corresponding to **some entries** in the “**Parameter Name**” column (e.g., “Normalization_method”, “Reduction_Method”, “AllMarkerGenes_Minimum_pct” and “AllMarkerGenes_Minimum_diff_pct”) are **automatically populated** with parameters predefined by PICASSA. Although these parameters are not manually selected by the user, they are important to report in the Materials and Methods section of a publication.

2. Upon downloading the report, the “**Next step**” button is enabled, allowing the user to proceed to the final step of Phase 1.

11.Check

1. Click “**Check**” to access a **table listing** any **unsaved items**.
2. After reviewing the table, you can **return** to the **corresponding step** to **save** them and then click “**Check**” again to verify completion, as the table **updates** dynamically to reflect the current status.

Note: Users are encouraged to save all data before proceeding.

3. Choose one of the following **options**:
 - i. Continue to Phase 2 via the “**Open Phase 2**” button.
 - ii. Return to PICASSA home page via the “**Return to Home**” button.
 - iii. End the current session via the “**End Session**” button.

PHASE 2

Phase 2 is an **exploratory** workspace designed to support the final **biological interpretation** of processed single-cell RNA-sequencing datasets, with **optional dataset refinement** available when further adjustment of the data is needed before final interpretation.

Unlike Phase 1, which follows a structured step-by-step workflow, Phase 2 is intentionally **modular**. The available analyses are organized into interconnected tabs that can be accessed in any order, allowing users to move between analyses based on the biological questions being investigated and the results emerging during the analysis.

Analysis typically begins in the “**Clustering**” tab, where the active dimensionality reduction plot and cluster structure are displayed. Users can then analyze gene expression by using the corresponding tabs to explore marker genes, visualize selected gene expression, apply annotation tools, and perform cluster-specific analyses such as differential expression and downstream enrichment analysis, to **interpret** cell populations. Inspection of these results may reveal contaminating populations, poorly resolved clusters, clusters containing mixed cell types, or biologically related clusters. In such cases, users can return to the “**Clustering**” tab to **refine** the dataset by removing clusters, performing subclustering, or merging populations. After refinement, the updated seurat object can be downloaded and reused in later Phase 2 sessions, enabling annotation and inspection to continue in subsequent Phase 2 sessions using the improved data.

Phase 2 should therefore be viewed as an **iterative cycle** of interpretation, refinement when required, and re-analysis rather than a linear sequence of steps. Users may repeatedly interpret cluster identities, refine cluster structure, and reassess results until a satisfactory cellular landscape has been established. Once the **final dimensionality reduction**, **clustering**, and cell-type **annotations** have been defined, users can proceed with **downstream biological analyses** including evaluation of cluster abundances, assessment of specific gene-expression patterns, differential expression analysis, gene set enrichment analysis (GSEA), over representation analysis (ORA), drug-gene interaction analysis, trajectory inference, and cell-cell communication analysis to generate biological conclusions from the finalized dataset.

This chapter of the manual provides an overview of the functionality of each analysis tab to support biological interpretation and guide conclusions from your scRNAseq dataset.

IMPORTANT NOTE ON NOMENCLATURE: Phase 2 can be initiated using either a seurat object generated by PICASSA in Phase 1 or a seurat object created externally. PICASSA generated objects use UMAP by default, whereas externally generated objects may contain UMAP, t-SNE, or both. If both are available, the user can choose which visualization to display. Although both options are supported, UMAP is generally recommended because it serves as the basis for several downstream analyses and visualizations, for instance trajectory inference with Monocle 3. To avoid repeatedly referring to specific dimensionality reduction methods, the term **Dimensionality Reduction Plot (DRP)** will be used throughout Phase 2 to refer to the active UMAP or t-SNE visualization displayed in the interface.

Get Started

If you already have a `seurat.rds` object to explore:

1. Under “**A) Start with your data**”, **browse** and **load** the `seurat.rds` file. Loading is complete once the message “**Upload complete**” appears.
Note: Loading can take **some time**.
2. Select the **organism** used in the experiment.
3. **Indicate** whether the `seurat.rds` object was **created with PICASSA or not**.
4. Click “**Run**” to load the data and access the different **downstream analysis** module tabs.

Note: If the object **was not created in PICASSA Phase 1**, a **conversion** step is performed in the “**Data table**” tab to ensure **compatibility** with the application. During this step, assign to each required field in the sidebar (e.g., cluster identities, group labels, sample information, mitochondrial/ribosomal percentages, and cell cycle scores) the corresponding metadata column in the table on the right by selecting an option using the dropdown menus, which are automatically populated with the metadata columns from the table. If a corresponding metadata column is not available, select “**No Column**”. Once all required fields have been specified, the “**Convert**” button will become available; click it to proceed. Once the conversion is completed, the user is asked to click “**Download seurat object (PICASSA format)**”, which activates the “**Next step**” button to access the remaining **downstream analysis** module tabs.

Alternatively, if you want to explore Phase 2 without using your own data, you can use an example dataset. Under, “**B) Start with demo dataset**” choose:

- “**Load example**” to **automatically load** the **input files** and populate the **required input information**; then click “**Run**” to proceed to the Phase 2 workflow.
- “**Download example data**”, to download the demo `.rds` file and **manually test file selection, uploads, and completion** of the **required input information**, and optionally continue through Phase 2 workflow.

Clustering

When starting Phase 2 workflow, the “**Clustering**” tab displays the initially loaded **active DRP**, together with its current **clustering configuration** and the **number of cells in each cluster**. These are derived either from the `.rds` `seurat` object downloaded in Step 9 of Phase 1, or from an object created externally to PICASSA.

As described in the introduction to the Phase 2 chapter, the interpretation and refinement of the clustering structure is usually an iterative process. Unless the marker genes generated in Phase 1 allow users to confidently interpret the DRP and consider the current cellular landscape sufficiently informative for their analysis, users will generally first use the subsequent Phase 2 tabs to further explore gene expression and determine whether refinement of the active DRP is needed. If no refinement is required, users can leave the clustering unchanged; in that case, all Phase 2 tabs will continue to use the initially loaded active DRP and clustering configuration. If refinement is needed, this tab provides the

options to **refine the cluster structure**. The resulting active cluster structure is then used across all other Phase 2 tabs.

The following options are available in the sidebar to refine the active DRP according to the user needs:

- **Subclustering:** One or more clusters can be chosen in the “**Clusters to divide**” field together with a “**Resolution**” value. Clicking “**Subcluster**” divides the selected cluster(s) from the active DRP into smaller groups, which are subsequently displayed as updated clusters in the active DRP, based on similarities in cell gene expression patterns. In other words, this option examines the gene expression profiles of the cells within the selected cluster, identifies shared patterns, and uses those similarities to create subclusters. Higher resolution values generally produce a greater number of resulting subclusters.

Note: A prefilled “**Resolution**” value of 0.1 is automatically set as initial guidance. If smaller resolution values result in no or insufficient subclustering, users should explore a higher value within the accepted PICASSA range of 0.1 to 2.

Note: One effective way to reveal hidden **heterogeneity** within a cluster is, for example, to visualize gene expression using **feature plots** as it allows gene expression to be inspected directly across the DRP and can reveal spatially distinct expression patterns within the same cluster. If different regions within the cluster express distinct informative genes, this may suggest that **subclustering** could reveal **biologically meaningful subpopulations**.

- “**Merge and/or rename clusters**”: The “**Old**” column contains the current cluster labels, while the **New** column is editable and defines the cluster labels that will be applied after renaming or merging.
 - **Merge clusters:** **Locate** the clusters you want to merge in the “**Old**” column and **assign** them the **same label** (numeric or text) in the corresponding cells of the “**New**” column. For example, assigning the value 5 to both clusters 5 and 6 of the active DRP will merge them into a single cluster labeled 5.

Note: Merging clusters can be useful, for example, when several clusters represent closely related cell populations that should be analyzed together. For instance, clustering may separate macrophages into multiple subclusters reflecting different activation states or transcriptional programs, but if the goal is to treat all macrophages as one broader cell population, these clusters can be assigned the same label and merged.
 - **Rename cluster(s):** **Locate** the cluster of the active DRP you want to rename in the “**Old**” column and **replace** the corresponding value in the “**New**” column with a custom label (e.g., Macrophages, T cells). The cluster will be renamed accordingly.

Note: Renaming clusters corresponds to **cluster annotation**, where clusters are assigned **biologically meaningful labels** based on their inferred cell identity, as determined through gene expression-based analyses.

After assigning the desired values or labels in the “**New**” column, click “**Merge and/or rename clusters**” to apply the selected merge and/or rename changes to the active DRP.

- “**Remove clusters**”: Clusters can be chosen in the “**Clusters to remove**” field and discarded from the active DRP by clicking “**Remove clusters**”. These clusters are then excluded from the dataset and from the subsequent analyses.

Note: Removing clusters differs fundamentally from merging, renaming, or subclustering because cells are permanently excluded from the dataset and can alter the composition of the dataset. As a result, several preprocessing steps must be repeated to ensure that the remaining cells are analyzed in the correct biological context. Accordingly, following cluster removal, PICASSA initiates a short, temporary stepwise workflow under the **"Final Results"** tab that is analogous to the **Phase 1** preprocessing workflow, although doublet detection is not included, as this quality control step is based on the initial number of loaded cells and should therefore be performed on the original dataset. During this workflow, the remaining data are reprocessed, including cell and gene filtering, normalization, optional regression, dimensionality reduction, and marker gene identification, resulting in an updated seurat object. If this additional **cell filtering** is **not desired**, the quality-control thresholds can be adjusted to retain all remaining cells as explained under **"Read data + cell filtering"** in chapter 1. These preprocessing steps are important because the removal of cells can change the relationships among the remaining cell populations, influencing normalization, dimensionality reduction, clustering structure, and marker gene identification. For example, the removal of contaminating or biologically distinct cell populations may reveal transcriptional differences that were previously masked, leading to improved clustering and more informative marker genes. Similarly, users may intentionally retain only a subset of biologically related cells (e.g., macrophages) to perform a more detailed analysis. By focusing on a narrower cellular compartment, the updated DRP and clustering can reveal subpopulations that were not apparent when the cells were analyzed together with more divergent cell types. This is the only situation in Phase 2 that requires a temporary stepwise workflow. Once this workflow has been completed, all downstream modules can again be used independently of one another. The updated seurat object becomes available for download from the **"Save some data"** subpanel within the **"Marker genes"** panel of the **"Final results"** tab. Because the recalculated object may be computationally intensive to generate and require a lot of memory, it is recommended to download and reload the updated seurat object before continuing with downstream analyses. Importantly, further subclustering and/or merging/renaming can only be performed after the updated seurat object has been reloaded.

- **Refresh to retrieve original UMAP:** Select this option and click **"Run"** to discard any of the clustering modifications described above that have been applied, and restore the initially loaded DRP and clustering configuration.

Note: This option becomes available only after the clustering has been modified using at least one of the actions described above.

Note: If clusters are merged, renamed, or split, the updated seurat object can be downloaded directly from the **"Clustering"** tab. If clusters are removed, the updated seurat object can be downloaded from **"Save some data"** within the **"Marker genes"** panel of the **"Final Results"** tab, once the removal workflow has been completed.

Final results

This module mirrors the functionality of **Phase 1 – "9. Final Plots / Marker Genes"**. It is provided to allow users to simply reproduce results from Phase 1, when the final Phase 1 cellular landscape is sufficiently informative for their analysis and no refinement of the cluster

configuration was performed, making these outputs readily available in Phase 2. Alternatively, if the clustering structure was modified in Phase 2, this module allows users to regenerate the final outputs, ensuring that results reflect the updated dataset. In this case, it represents the final stage of preprocessing, enabling users to obtain the updated clustering alongside relevant summary information and marker gene results.

Expression analysis and cell enrichment

This panel begins with the option to modify clusters using the same functionality provided in the “**Clustering**” panel. This enables more refined exploration of the data, supporting improved cluster annotation, identification of potential contaminants, and more accurate interpretation of results. The panel includes three subpanels for downstream expression and enrichment analyses. Across these modules, multiple options are available for selecting genes or gene sets. However, the two common options shared across the subpanels are predefined [Gene Ontology \(GO\)](#) and [Molecular Signatures Database \(MSigDB\)](#) pathways. Depending on the selected subpanel, additional input options are also available, including manual specification of genes or upload of custom gene lists. Once the desired cluster modifications have been defined, the following subpanels become available for further analysis:

Gene expression

In this module the user specifies the genes of interest, either by selecting predefined GO and MSigDB pathways or by manually providing gene names. To facilitate clear labeling and organization of the resulting plots, the user is also required to assign a custom name to the selected gene set. The output of this module is a table along with some plots to help the user see results clearer:

- **Data info:** a **table** shows the total number of input genes, the number of genes found in the dataset and the genes not detected in the dataset. This helps assess coverage and ensures correct gene matching.
- **Feature plots:** visualizes spatial expression of the selected gene on the dimensionality reduction plot.
- **Violin plots:** shows the expression of the selected gene **per condition, in conditions within clusters and per cluster**.
- **Dot plot:** summarizes gene expression across clusters, where dot size represents the percentage of expressing cells and color indicates the average expression level.
- **Heatmap:** visualizes expression patterns across selected clusters and/or conditions, enabling a more global overview of gene behavior.

Average expression

This module enables users to explore the average expression of selected gene sets across clusters and conditions. The user defines the gene sets of interest using one of the following approaches: (i) choosing predefined GO pathways, (ii) choosing predefined MSigDB pathways, (iii) manually selecting genes, or (iii) uploading a file containing known marker genes. For the upload option, the file must be in .xlsx format, where each column represents a gene set or cell type and contains genes in Ensembl ID format, see example below:

B Cells	T Cells
ENSMUSG000000026395	ENSMUSG000000026395
ENSMUSG000000030724	ENSMUSG000000032093
ENSMUSG000000003379	ENSMUSG000000032094
ENSMUSG000000040592	ENSMUSG000000002033
ENSMUSG000000030577	ENSMUSG000000026012

To facilitate the required format, PICASSA provides a helper app called “**Gene Conversion**”, which enables users to convert gene names or orthologs (e.g., human or mouse) into the appropriate Ensembl IDs. The output of this section is similar to the previous one where a table along with some plots are generated for an easier interpretation of the results:

- **Data info:** a table, similar to the one in the “**Gene Expression**” module, to display the total number of genes loaded, the number of genes found in the dataset, and those not detected.
- **Average Feature Plots:** displays the spatial distribution of the average expression of each gene set on the dimensionality reduction plot.
- **Average Violin Plots:** show the distribution of average expression values across **clusters or conditions**, with a very important option to perform statistical comparisons using Mann U Whitney test.
- **Average Dot Plots:** summarizes the average expression across clusters, where dot size reflects the proportion of expressing cells and color represents the average expression level.

Single cell enrichment analysis

This step computes **gene set enrichment scores at the single-cell level**, allowing the activity of biological pathways or functional gene sets to be evaluated across individual cells and cell clusters. Similarly to the above process, the user selects the database or source to define the gene sets for enrichment analysis. The options are: (i) GO pathways (ii) MSigDB pathways and (iii) upload a custom gene set file as done in the “**Average Expression**” section. Then the user selects the enrichment method which is an algorithm used to calculate enrichment scores for each cell. Available methods include:

- **UCell:** Computes enrichment scores using a **rank-based Mann-Whitney U statistic**. Genes are ranked per cell, and the enrichment score reflects whether genes from the selected gene set are consistently ranked higher than background genes. UCell is optimized for sparse single-cell datasets and is typically the **fastest method**.
- **AUCell:** Evaluates whether genes from a gene set are enriched among the **top-ranked genes in each cell** by calculating the **Area Under the Curve (AUC)**. This method is widely used in single-cell regulatory network analysis.
- **ssGSEA - Single-sample Gene Set Enrichment Analysis:** calculates enrichment scores based on **cumulative distribution functions of ranked gene expression**

values. This approach is widely used but may be slower for very large single-cell datasets.

Each cell receives an enrichment score for each selected gene set. These scores can be visualized through several output tabs, similar to “**Gene Expression**” and “**Average Expression**” panels:

- **Data info:** a table summarizing how the selected gene sets map to the genes present in the dataset. For each gene set, the table reports the name of the pathway or gene set, the total number of genes originally defined in the pathway, the number of genes from the pathway that are present in the dataset and finally the gene names from the pathway that were not detected in the dataset.
- **Feature plots:** shows the average expression scores of the selected geneset.
- **Violin plots:** displays the distribution of average expression scores across **clusters**, **conditions** and **conditions in clusters**.
- **Heatmap:** visualizes the enrichment score of the selected gene set across the whole dataset. Specifically, this heatmap summarizes the average enrichment score of each gene set across clusters. Rows represent gene sets, while columns represent clusters. This visualization allows users to compare pathway activity patterns between gene sets across cell populations. Heatmaps provide a global overview of pathway activity across the dataset.

Annotation tools

This panel provides automated cell type annotation methods to assist with the biological interpretation of clusters identified during the analysis. The user must first select the desired annotation tool together with the corresponding tissue type, as annotation accuracy depends strongly on tissue specific marker genes. These methods assign biologically meaningful cell type labels to clusters by expression profiles or cluster associated marker genes against curated reference tissue type databases. The available annotation methods are the following: comparing cluster

- **ScType:** a fully automated platform for cell-type identification that enables accurate and ultra-fast annotation using the already preprocessed scRNA-seq expression matrix together with the comprehensive [ScType marker database](#), without requiring an external reference atlas. It is an unsupervised method that leverages marker-gene specificity across both cell clusters and cell types to identify highly specific positive markers, while also incorporating negative markers to provide evidence against assignment to a particular cell type and improve annotation selectivity.
- **scMayoMap:** an easy-to-use and accurate cluster based annotation method that uses the cluster marker genes as input and matches those genes against the tissue-specific [scMayoMapDatabase \(Additional File 1\)](#). For this reason, scMayoMap is **available** only after “**Final results**” has been run, as this module produces the marker genes used for annotation. It scores candidate cell types using both differential expression strength and the fraction of cells expressing each marker.

For every annotation method, a **score table** with cell type annotation scores and a set of plots for visualization are produced. Regardless of the annotation method used, each run will generate a dimensionality reduction plot showing the predicted annotated clusters. This provides a visual overview of how cells have been grouped and labelled by the method.

Each method also produces a second, method-specific plot to further support interpretation of the results:

- **ScType results:** The ScType **scores table** summarizes the top predicted annotation for each cluster. It includes the cluster identifier, the predicted cell type, the annotation score, the total number of cells in the cluster, the value QuarterOfNcells, and whether the prediction passes the ScType threshold (PASS). QuarterOfNcells is calculated as 25% of the number of cells in the cluster and is used as the internal threshold for accepting the annotation. For example, if a cluster contains 80 cells, then QuarterOfNcells = $80 \times 0.25 = 20$. A score of 18 would therefore not pass, whereas a score of 26 would pass. Predictions with scores below this threshold are marked as PASS = No and are reassigned to Unknown, indicating that the proposed label is not sufficiently supported for that cluster. ScType also produces a **bar plot** of annotation scores. In this plot, the x-axis shows the clusters, the y-axis shows the ScType scores, the fill color indicates the predicted cell type, and the label above each bar shows whether the score passed the threshold (Yes or No). This helps the user quickly identify which cluster labels are strongly supported and which should be interpreted more cautiously.
- **scMayoMap results:** The scMayoMap score result is typically saved in two complementary tables. The **main table** reports the cluster identifier together with the top predicted cell type or cell types, the normalized estimated score (ES.norm), and the matched marker genes supporting that prediction. ES.norm reflects the normalized strength of the match between the cluster marker profile and the reference marker set for the predicted cell type, with higher values indicating stronger support. The genes column does not list all genes in the dataset; rather, it lists the marker genes retained by scMayoMap as supporting evidence for the proposed annotation of that cluster. These genes help the user understand which markers contributed to the assigned label and whether the prediction is biologically plausible. The **second table** provides the score details for the candidate annotations. In addition to ES.norm, it includes ES.raw, which is the raw estimated score, and ES.se, which is the associated standard error. Together, these values help the user evaluate the strength and stability of the predicted annotation. If more than one candidate cell type is reported for a cluster, this suggests that the cluster shares marker expression with multiple related populations and may therefore require closer inspection. Along with the annotated dimensionality reduction plot, scMayoMap also produces a **dot plot** of annotated cell types. This plot visualizes the expression of the relevant marker genes across clusters to help confirm and interpret the assigned labels. In general, dot size reflects the proportion of cells expressing a marker, while color reflects the expression level, allowing rapid assessment of how strongly and how broadly the supporting markers are expressed within each cluster.

Projections

Projection analysis allows you to load a reference seurat object, representing a known single-cell RNA-seq experiment using its .rds file. This reference can help interpret and define clusters in your current dataset, especially if the experiments are similar.

To perform a projection analysis:

1. **Load reference seurat object**

Select a seurat object in .rds format to load and use for projecting clusters.

2. **Set Log2 Fold Change Thresholds**

Define the Log2FC thresholds for identifying differentially expressed genes (DEGs) in both the current and reference datasets.

3. **Set minimum percentage thresholds**

Specify the minimum percentage of cells expressing a gene in each group, used to detect DEGs.

4. **Set adjusted p-value thresholds**

Define the adjusted p-value thresholds to determine statistical significance for DEGs in both datasets.

Once the projection is complete, a series of visual outputs are generated to help interpret how the datasets relate to one another. The outputs depend on the direction of the projection: The first two are from the same method and the last one is from the method that Seurat uses.

- **For the first one**, basically the goal is to map clusters from one seurat object (reference or data) onto another (data or reference) by comparing their differential gene expression profiles, quantify the similarity and produce visual summaries. For each cluster pair, it finds overlapping marker genes and computes two scores: a) Set Overlap Score (S): based on proportion of shared genes and b) Expression Score (E): based on shared expression magnitudes. Taking these 2 scores, a final score is then produced, by simply multiplying them together.
- The Seurat method **projects reference data onto a query dataset**, it uses a process called **reference mapping**, which is designed to transfer cell type labels, embeddings, or other annotations from a well-annotated reference to a new, possibly unlabelled query dataset.

When Your Data is Projected onto the Reference Data

- **Table**
The table summarizes pairwise comparisons between clusters from your data and the reference dataset (represented by the cluster_1 and cluster_2 columns, respectively), quantifying the overlap between their marker genes. For each cluster pair, it reports overlap-based metrics (S score, E score, and the combined Overlap Score), along with the number of shared marker genes and the total number of marker genes in the reference cluster.
- **Alluvial Plot**
Displays the top five genes per cluster. Highlighted lines represent the genes with the maximum log2 fold change (Log2FC) for each cluster compared to the reference.
- **Violin Plot**
A violin plot shows the average expression of genes from your data's clusters, projected onto the clusters of the reference dataset shown on the x-axis.
- **Feature Plot**
A feature plot displays the expression of genes from your data's clusters overlaid on the reference dataset's UMAP or spatial layout.

When the Reference Data is Projected onto Your Data

- **Table**
The table summarizes pairwise comparisons between clusters from the reference dataset and your data (represented by the cluster_1 and cluster_2 columns,

respectively), quantifying the overlap between their marker genes. For each cluster pair, it reports overlap-based metrics (S score, E score, and the combined Overlap Score), along with the number of shared marker genes and the total number of marker genes in the reference cluster.

- **Alluvial Plot**

Focuses on the top five genes for each cluster, with highlighted lines indicating those with the highest Log2FC when projected onto your data.

- **Violin Plot**

A violin plot showing the average expression of genes from the reference clusters, projected onto the clusters in your dataset shown on the x-axis.

- **Feature Plot**

A feature plot visualizing the expression of genes from the reference clusters, mapped onto your data's UMAP or spatial representation.

When Seurat method is used

- A table showing a summary of the prediction scores transferred from the reference to the query dataset, these are the scores Seurat assigns when it maps a cell to a specific identity.
- Generates a UMAP plot of the query dataset, colored by the transferred annotations.
- Creates a bar plot showing the mean prediction score for each possible label.
- Creates a violin plot of prediction scores across all cells for each label.
- Generates a bar plot showing the number of cells assigned to each predicted identity.

Differential Expression Analysis (DEA)

This module allows comparison of gene expression, **between conditions across all data, between conditions within each cluster, between specific clusters and between specific conditions from a specific cluster**. In all cases, the analysis returns a table of differentially expressed genes (DEGs), along with summary plots such as volcano and violin with boxplot overlay.

Between-Conditions across all data

This section performs differential expression analysis between user-defined conditions across the entire dataset. Although similar comparisons can be generated in the “**Final Results**” tab, those analyses are based on the initial selections provided earlier in the workflow. In contrast, this section allows users to define custom comparisons, providing greater flexibility for exploring condition-specific differences.

- **Select the number of comparisons:**

Users first specify the number of comparisons they wish to perform. Each comparison represents a differential expression test between selected conditions.

Examples:

- Condition 1 vs Condition 2
- Condition 1 vs Condition 2 + Condition 3

In this case, the total number of comparisons would be two.

- **Define Comparison Groups**

For each comparison, users must define the conditions that will be included in the two groups being compared. This allows flexible grouping of experimental conditions when testing for differential expression.

Between-Condition Comparison Within Each Cluster

This analysis compares different conditions (e.g., treatment groups) **within each selected cluster**.

- **Select Clusters**

Choose the clusters in which you want to perform the condition comparisons.

- **Select Condition Comparisons**

Define which conditions to compare within each cluster.

Between-Specific Clusters

This analysis compares different clusters.

- **Specify the Number of Comparisons**

Enter the total number of comparisons you want to perform.

Example:

- Cluster 1 vs. Cluster 2
- Cluster 1 and 3 vs. Cluster 2

In this case, the number of comparisons is **2**.

- **Define Clusters to Compare**

Manually input the clusters involved in each comparison.

Between-Specific conditions from a specific cluster

- **Specify the Number of Comparisons** Enter how many unique comparisons you want to perform.

Example:

- Cluster 1, Condition 1 vs. Cluster 2, Condition 1
- Cluster 1, Condition 1 vs. Cluster 2, Condition 2 and Cluster 3, Condition 2

In this case, the number of comparisons is **2**.

- **Define the Comparisons**

Input the exact cluster-condition groupings you wish to compare.

Expression Parameters that apply to all DEA subtabs:

- **Average Log2 Fold Change (log2FC) Threshold**

Set the minimum log2 fold change required to consider a gene differentially expressed. Note: A log2FC value of |1| corresponds to a two-fold change in expression (i.e., FC = 2 or 0.5).

Fold change (FC) is calculated as: $FC = 2^{\log_2FC}$ Example: $\log_2FC = 1 \rightarrow FC = 2$, $\log_2FC = -1 \rightarrow FC = 0.5$

- **Minimum Percentage Threshold**

Specify the minimum proportion of cells expressing the gene in either group (e.g., 25%).

- **Minimum Percentage Difference**

Specify the minimum proportion difference of cells expressing the gene between the first and second group of comparison (e.g. 10% = 0.1).

- **Adjusted P-Value Threshold**

Set the significance level for identifying differentially expressed genes, corrected for multiple testing.

Output that apply to all DEA subtabs

- **Table**
A table listing all identified differentially expressed genes (DEGs) for each comparison.
- **Volcano Plot**
One volcano plot per comparison, displaying the log2 fold change and adjusted p-values for each gene.
 - The plot title includes the number of DEGs detected.
 - The shape of each data point indicates whether the gene meets the **Minimum Percentage threshold**.
- **Violin Plot**
A combined violin and boxplot visualizing the expression distribution of a selected gene across the chosen comparisons.

Enrichment Analysis (ORA)

This step performs **Over-Representation Analysis (ORA)** to identify biological pathways or functional categories that are significantly enriched in a set of **differentially expressed genes (DEGs)**. The analysis can be repeated multiple times using different parameters or gene sets.

1. **Select a DEA Result**
Choose a previously generated **differential expression analysis (DEA)** result.
2. Any DEA result produced during the workflow can be selected, including those generated in the **Final Results** section.
3. **Choose a Database for Enrichment**
Choose the biological database used to define gene sets for the enrichment analysis. Available databases include:
 - **Gene Ontology (GO): BP (Biological Process)**: biological pathways and cellular processes. **MF (Molecular Function)**: molecular activities of gene products. **CC (Cellular Component)**: cellular locations of gene products. **ALL**: combines all GO categories
 - **MSigDB Collections (C1-C8, H)**: Curated geneset collections containing pathways, regulatory signatures, and experimentally derived gene sets.
 - **KEGG**: Pathway based gene sets describing metabolic and signaling pathways.

Note: that gene sets originating from different databases may vary in their gene coverage and curation. As a result, enrichment results may differ depending on the selected database.

4. **Select a P-Value Adjustment Method**
Choose a method to correct for multiple hypothesis testing:
 - holm**: Holm's correction (step-down Bonferroni)
 - hochberg**: Hochberg's correction (step-up Bonferroni)
 - bonferroni**: Very strict Bonferroni correction
 - BH** (default): Benjamini-Hochberg correction (commonly used; balances sensitivity and specificity)
 - BY**: Benjamini-Yekutieli correction (more conservative than BH)
 - fdr**: False Discovery Rate correction (equivalent to BH)
 - none**: No adjustment

The **Benjamini-Hochberg (BH)** method is commonly used because it balances sensitivity and specificity while controlling the false discovery rate.

5. Set Adjusted P-Value Threshold

Specify the cutoff for statistical significance (e.g., **0.05**, which is commonly used).

The following results are generated for each ORA run:

Results – Tables:

- **RDS files** containing the complete results of the ORA analysis are generated to facilitate reproducibility and further downstream analysis in R. These objects allow users to reload and continue their analysis without repeating Phase 2, and can also be used with the companion tool Pathway Plotter for re-visualization of the results.
- **Show results:** Select whether to display the results for “**Up- and Downregulated Genes**” or “**All Filtered Genes**”.
 1. **Results table:** Enrichment result table.
 2. **Frequency and bar plot:** If results for `Up- and Downregulated Genes` was chosen then the option to show “**Up**” or “**Down**” becomes available.
 1. **Frequency table:** Shows how many genes from the comparison were found in each enriched pathway.
 2. **Bar plot:** A bar plot visualizing the number of genes found in each enriched pathway (based on the frequency table above).

Main plots:

- **Dot Plot**

By default it displays the top 20 significant pathways where **y-axis** are the pathway names, **x-axis** is GeneRatio value, **dot color** displays the adjusted p-value and **dot size** the number of genes found in the pathway.
- **Enrichment map**

Displays the pathways shown in the dot plot above and their relationships based on shared genes. Where **dot color** is the adjusted p-value and **dot size** is the number of genes in each pathway
- **Gene network**

By default it visualizes the top 3 most significant pathways and the genes they contain where **dot color (gene)** shows the expression value **and the dot size (pathway)** the number of associated genes.
- **Dot Plot - up/down**

Dot plot showing the top 10 significant pathways enriched **within each expression** (up or downregulated DEGs).
- **Enrichment map - up/down**

Similar to the main enrichment map, but grouped **per expression**, visualizing pathway relationships and shared genes.
- **Gene network - up/down**

Shows the top 3 significant pathways **per expression**, with gene symbols included.

Gene Set Enrichment Analysis (GSEA)

This module performs **Gene Set Enrichment Analysis (GSEA)** to identify biological pathways that are significantly enriched at the top or bottom of a **ranked gene list** derived from differential expression results. Unlike Over-Representation Analysis (ORA), which uses a predefined list of significant genes, GSEA considers the **entire ranked gene list**, allowing the detection of subtle but coordinated changes in gene expression. The analysis can be repeated multiple times using different parameters or gene set sources.

- **Select a DEA Result**

Choose a previously generated **differential expression analysis (DEA)** result. Any DEA result produced during the workflow can be used, including those generated in the **Final Results** section.

- **Choose a Category for Analysis:** Choose the gene set collection used for the enrichment analysis. Two options are available:

- **MSigDB Gene Sets:** Curated geneset collections from the **Molecular Signatures Database (MSigDB)**. These collections include pathways, regulatory targets, biological processes, and hallmark gene signatures. Available categories include:

C1-C8: curated geneset collections covering biological processes, regulatory signatures, and pathway databases

H (Hallmark): concise gene sets representing well-defined biological states or processes. [More information on the available categories \(MSigDB\)](#)

- **Custom Gene Sets**

Users may upload their own gene sets in **Excel (.xlsx) format**. The file should be structured such that: each column represents a gene set (e.g., a pathway, module, or signature) and each row contains genes belonging to that gene set. Example:

B Cells

ENSMUSG00000026395

ENSMUSG00000030724

ENSMUSG00000003379

T Cells

ENSMUSG00000026395

ENSMUSG00000032093

ENSMUSG00000032094

- **Gene identifier type:**

The gene identifiers used in the uploaded geneset file must match the identifier format used in the dataset. Supported formats include: **Ensembl IDs and Gene symbols**. Ensembl IDs are generally recommended because they are stable identifiers and reduce ambiguity caused by gene symbol synonyms. If the seurat object was generated during **Phase 1 of PICASSA**, gene identifiers are typically expected to be **Ensembl IDs**. If an external seurat object is used instead, the identifiers in the uploaded geneset file must match the gene identifiers present in that dataset. Gene identifiers can be converted to **Ensembl IDs** using the **Gene Conversion** tool available in PICASSA.

- **Set Adjusted P-Value Threshold**

Define the significance cutoff (commonly set at **0.05**).

- **Choose a Ranking Metric**

Select how genes should be ranked for GSEA:

- **Average Log2FC:** Commonly used and stable
- **Rank Metric:** Combines the direction of differential expression with statistical significance using the metric = $\text{sign}(\log_2\text{FC}) \times -\log_{10}(\text{p-value})$.

The following results are generated for each GSEA run:

Results – Tables:

- **RDS files:** To facilitate reproducibility and further analysis, the complete results of each GSEA run are stored as RDS files, allowing the analysis to be reloaded without re-executing the workflow. These files can also be used in downstream applications, including the Pathway Plotter tool, for additional visualization and exploration.
- **Results**
Lists the pathways that are significantly enriched at either end of the ranked gene list.
- **Frequency Table**
Summarizes how frequently each gene appears across the enriched pathways. The percentage indicates the proportion of enriched pathways containing that gene. For example, a value of 50% means that the gene was present in half of the enriched pathways.
- **Barplot**
Visualizes the frequency of genes across the enriched pathways. Bar height represents the percentage of enriched pathways containing each gene, helping identify genes that are recurrently associated with the enrichment results.

Main plots:

- **Dot plot**
By default, the plot displays the top 20 significant pathways ranked by the selected variable, with the Normalized Enrichment Score (NES) used as the default x-axis variable. Users can choose alternative variables and manually select specific pathways to display. The y-axis shows pathway names, dot color represents the adjusted p-value, and dot size indicates the number of genes in the pathway.
- **Enrichment map**
Visualizes relationships between the pathways currently displayed in the dot plot based on shared genes. Dot color represents the adjusted p-value, while dot size indicates the number of genes associated with each pathway.
- **Gene network**
Shows pathways and their associated genes. By default, the three most significant pathways are displayed. Gene color represents the expression value, while pathway size indicates the number of genes in each pathway.
- **Enrichment plot**
Visualizes the **Running Enrichment Score** (y-axis) across the **Rank in Ordered Dataset** (x-axis). It also shows the **Ranked List Metric** below, to view enrichment for multiple pathways to determine whether they are enriched at the top or bottom of the ranked gene list.

Drug-Gene Interaction Analysis

This analysis identifies potential drug-gene interactions using DEA results and the [Drug-Gene Interaction Database \(DGIdb\)](#).

1. Select a DEA Result

Choose any differential expression analysis result previously generated, including those from the Final UMAPs section.

The **Drug-Gene Interaction Database (DGIdb)** is a comprehensive resource that curates known drug-gene interactions and potential druggable targets. It integrates data from multiple sources, including peer-reviewed literature, public databases, and expert curation. This makes it a powerful tool for supporting drug discovery and precision medicine.

[Explore the DGIdb](#)

This panel produces a table and a network plot.

- **Results Table**
Displays the interactions found between selected genes and known drugs.
- **Network Plot**
A visual representation of the drug-gene interaction network where **red nodes** represent drug names **blue nodes** represent gene names and **edge width** corresponds to the interaction score between the drug and gene

Trajectory analysis (Monocle3)

Trajectory analysis allows the study of cell lineage progression by ordering cells along a pseudotime axis based on their transcriptomic similarity. This approach helps identify differentiation pathways, lineage branch points and dynamic gene expression changes during cellular transitions. The trajectory is inferred using Monocle3, which constructs a trajectory graph in a low-dimensional embedding (UMAP) and assigns a pseudotime value to each cell.

1. Select Clusters for Analysis

Users may optionally select specific clusters to include in the trajectory analysis, otherwise, all clusters in the dataset are used. Restricting the analysis to selected clusters can help focus on a particular lineage or biological process.

2. Select the Initial Cluster

The initial cluster defines the starting point of the trajectory (root of pseudotime) and should be selected based on prior biological knowledge, as cells from this cluster are used to initialize pseudotime ordering.

3. Run Trajectory by Condition

Trajectory analysis can optionally be performed separately for each experimental condition. This option is useful for comparing trajectory structures between groups.

4. Identify Genes Varying Along the Trajectory

Detect genes whose expression changes along the trajectory. This is done using the Geary's C spatial autocorrelation test, which measures how similar gene expression is among neighboring cells in the trajectory graph. Genes with strong spatial structure along the trajectory are considered trajectory-associated genes.

Trajectory analysis generates multiple outputs that are designed to characterize cell state transitions, lineage relationships, and gene expression dynamics along pseudotime:

Notes on trajectory visualization: Three types of nodes appear in the trajectory graph:

- **White circles:** Root nodes (starting points of pseudotime)
- **Black circles:** Branch points where differentiation paths diverge (lineage split points)
- **Light gray circles:** Leaf nodes representing terminal states (endpoints of differentiation)

Trajectory plots:

- **Cluster Trajectory Plot:** A UMAP showing the selected clusters together with the inferred trajectory graph connecting related cell states.
- **Pseudotime Trajectory Plot:** Cells are colored according to their pseudotime values, showing how cells progress along the trajectory.

- **Pseudotime with Branch Nodes:** A trajectory plot highlighting branch points, roots, and terminal states, providing insight into lineage branching events.
- **Pseudotime Density per Cluster:** A density plot showing the distribution of pseudotime values across clusters where **x-axis** shows the pseudotime and **y-axis** the cluster identities.

Top 5 trajectory genes

If “**Find genes that are differentially expressed across a single-cell trajectory?**”

is “**Yes**”, then the top five genes whose expression changes along the trajectory are displayed, along with a results table for all tested genes. Each gene is evaluated using the Geary’s C spatial autocorrelation statistic, which measures the similarity of gene expression among neighboring cells along the trajectory. Lower Geary’s C values indicate stronger spatial structure in gene expression, whereas higher values suggest weaker spatial patterns. Genes are ranked based on their statistical significance (q value) and their Geary’s C score, and the five most significant genes are visualized.

- **Gene Statistics Table:** Each row in the table represents one gene tested along the trajectory. The **status** column indicates whether the statistical test was successfully executed. The **p_value** column reports the p-value obtained from the Geary’s C spatial autocorrelation test, which evaluates whether gene expression shows significant spatial correlation among neighboring cells along the trajectory. The **geary_test_statistic** column shows the statistic used to compute the p-value, while **geary_C** represents the Geary’s C spatial autocorrelation score, where lower values indicate stronger spatial structure in gene expression. The **gene** column contains the gene identifier from the **seurat** object; if both gene symbol and Ensembl ID are available, additional columns display each identifier separately. The **q_value** column provides the adjusted p-value calculated using the Benjamini–Hochberg false discovery rate (FDR) correction and is used to rank genes by statistical significance. Genes with low q-values and low Geary’s C scores exhibit the strongest trajectory-associated expression patterns.
- **Feature plots of top 5 genes:** Five feature plots display the expression of the top trajectory associated genes across the UMAP embedding. These plots help identify where along the trajectory each gene is expressed.
- **Combined expression plot:** A single UMAP displays the expression of all five genes simultaneously. Expression values are scaled as %Max, meaning each gene’s expression is normalized relative to its maximum observed value. This allows easier comparison of expression patterns across genes.

Specific genes

Users can also enter custom gene names to visualize their expression dynamics along pseudotime. This allows tracking of genes of interest across the trajectory and identifying where they become activated or repressed during differentiation.

- **Genes by cluster:** Displays gene expression as a function of pseudotime, with cells colored according to their cluster assignment. This allows visualization of how expression patterns change across clusters along the trajectory.
- **Genes by pseudotime:** Displays gene expression as a function of pseudotime, with cells colored according to their pseudotime value. This allows visualization of expression trends along the inferred trajectory.

- **Genes on trajectory:** Displays gene expression projected onto the trajectory. Cells are colored according to relative gene expression, allowing identification of where along the trajectory the gene is most highly expressed.

LIANA: Ligand-Receptor Interaction Analysis

The [LIANA tutorial](#) describes LIANA as a framework for ligand-receptor analysis that decouples inference methods from interaction resources, so different scoring approaches can be combined with different prior-knowledge databases. In PICASSA, LIANA is used to infer putative cell-cell communication from scRNA-seq expression data together with curated ligand-receptor knowledge. The broader [LIANA+ paper](#) extends this framework to additional settings, including spatial and multi-omics analyses, but in PICASSA the focus is the ligand-receptor analysis workflow for clustered single-cell data.

1. **Choose annotation column:** Select which cell identity column will define the sender and receiver populations in the analysis. Available options are:
 - **final_clusters:** Cluster identities currently active in PICASSA.
 - **scType:** Annotations generated by ScType.
 - **scMayo:** Annotations generated by scMayoMap.
2. **Choose a resource:** Select the ligand-receptor knowledge resource used to define which ligand-receptor pairs are considered candidate interactions. In LIANA, a resource is the prior interaction database used in the analysis, whereas the method is the scoring strategy applied to those interactions. **PICASSA** provides **Consensus** as the **default** option. This is **LIANA's default** ligand-receptor resource, generated from the consensus of several expert-curated resources retrieved through OmniPath/OmnipathR and then refined using additional quality-control criteria, including literature support, complex handling, and localization information. Additional resources available in PICASSA include **Baccin2019**, **CellCall**, **CellChatDB**, **Cellinker**, **CellPhoneDB**, **CellTalkDB**, **connectomeDB2020**, **EMBRACE**, **Guide2Pharma**, **HPMR**, **ICELLNET**, **iTALK**, **Kirouac2010**, **LRdb**, **Ramilowski2015**, **OmniPath**, and **MouseConsensus**.
3. **Choose method(s):** Select one or more LIANA inference methods to score and rank candidate ligand-receptor interactions. In PICASSA, the available methods are:
 - **natmi:** LIANA implementation of the NATMI scoring framework.
 - **sca:** LIANA implementation of the SingleCellSignalR scoring framework.
 - **connectome:** LIANA implementation of the Connectome scoring framework.
 - **logfc:** A LIANA score inspired by iTALK based on 1-versus-rest log fold change.
 - **cellphonedb:** LIANA implementation of the CellPhoneDBv2 framework.

Note: In the LIANA tutorial, **natmi** and **sca** are among the default LIANA methods. When more than one method is selected, LIANA can aggregate results across methods to obtain consensus-ranked interactions rather than relying on a single scoring scheme.

LIANA panel generates a table and multiple interactive plots such as heatmap, dot plot and a chord diagram:

- **Data summary:** Table of ligand-receptor interactions, including source and target cell types, ligand and receptor complexes, method-specific scores, and the consensus aggregate_rank.

Note: The aggregate rank can be interpreted similarly to a p-value: lower values indicate interactions ranked consistently highly across methods.

- **Dot plot:** Displays ligand-receptor interactions between selected source and target cell types. Dot color represents the expression magnitude, while dot size indicates interaction specificity.

Note: Interactions not expressed in at least 10% of cells in both clusters are excluded from the dot plot.

- **Filtered plots:** These plots are generated from the LIANA interaction results after two consecutive filtering steps. First, when more than one LIANA method is selected, the results are aggregated and only highly concordant interactions are retained using **aggregate_rank ≤ 0.01** , following the LIANA tutorial recommendation. Second, in this panel, the user can further filter the retained interaction table by choosing a numeric column, a comparison rule, and a cutoff value. The comparison rule can be either **$\geq$ (greater/equal)** or **$\leq$ (less/equal)**, meaning that only interactions with values above or equal to, or below or equal to, the **selected threshold** in the **chosen column** are kept for visualization.
 - **Heatmap:** Displays the frequency of significant ligand-receptor interactions between source and target cell types. Higher values indicate more frequent communication events.
 - **Chord plot:** Visualizes communication frequencies between selected source and target cell types, with connecting ribbons indicating the relative strength of the inferred interactions.

Convert and download seurat object

This module is available for seurat objects generated in PICASSA Phase 1, where both gene symbols and Ensembl identifiers are retained. Its purpose is to standardize feature naming by converting all identifiers to gene symbols, ensuring compatibility with downstream analysis tools and external resources that primarily rely on gene symbol annotations.

During this process, features are systematically mapped from Ensembl IDs to their corresponding gene symbols, resulting in a unified and consistent feature set. This step is particularly useful for downstream analyses such as enrichment, visualization, and cross-tool integration, where mixed or duplicated identifiers may otherwise lead to inconsistencies or loss of information.

Please note that this process may take some time depending on the size of the dataset.

Report

The report can be downloaded at any time from the movable panel located in the top-right corner, allowing all performed steps to be saved.